ChronoTrace: Calibrating Residual Deleted Neighbor Inference After Node Deletion in Dynamic Trust Graphs

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Abstract

Trust and reputation platforms publish who rated whom, and data-protection law obliges them to erase a user's data on request. Erasure deletes a user's ratings, yet the counterparties' other ratings stay public, so the published graph may still reveal the user's financial and social relationships. Existing tools cannot measure this risk: link prediction needs the erased user's neighbors, unlearning certifies models rather than the published graph, and an attack's accuracy alone cannot separate the user's relationships from other users' popularity and timing. We propose ChronoTrace, a benchmark that pairs an attack with a calibration. The attack ranks every remaining user as a likely former neighbor, with or without knowledge of when the erased user was active, and trains only on erasures it simulates on the published graph. The calibration reruns the attack on reference graphs that keep every user's degree, or signed activity over time, but randomize partners, and with degree-matched controls attributes its accuracy to degree, timing, or what remains. On 60 erasures in each of three timestamped trust graphs (BitcoinOTC, BitcoinAlpha, Epinions), the attack without timing knowledge reaches an area under the ROC curve (AUC) of 0.87–0.93 (0.5 is chance), but degree explains all of it: among users of equal degree it performs at chance (degree-matched AUC 0.46–0.52). On the Bitcoin trading graphs, a window covering the user's active months adds signal beyond degree (degree-matched AUC 0.81), and on BitcoinOTC sequential user IDs let an adversary estimate this window from the published graph alone, gaining +0.018 AUC, about half the exact-window gain. Precision stays low: top-five precision is at most 0.19. ChronoTrace thus lets platforms audit an erasure before publishing and identifies two cheap fixes: hiding accounts left without ratings and not numbering users in order of first activity.

1 Introduction

Trust and reputation platforms make relationships public by design. Traders on Bitcoin OTC, a marketplace for over-the-counter bitcoin trades, rate each other after a trade on a scale from −10 to +10 so that newcomers can judge a counterparty's history [15], and members of Epinions, a consumer review site, declared which reviewers they trusted and which they distrusted [19]. Data-protection law entitles each of these users to have their data erased: Article 17 of the EU General Data Protection Regulation requires a platform to erase a user's personal data on request [9], and a platform honors such a request by deleting the account and every rating the user gave or received.

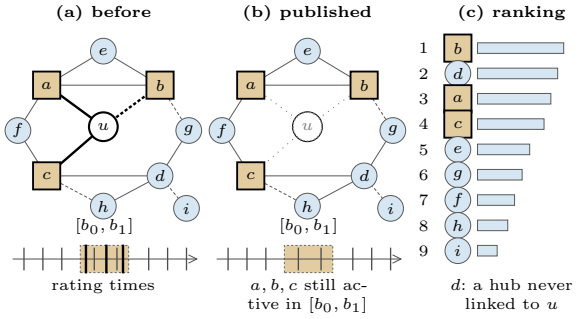


Figure 1: Running example. (a) Trader u traded with a, b, c (squares) in window $[b_0, b_1]$. (b) Erasure deletes u's ratings. (c) A ranking of the remaining users. Solid: positive rating; dashed: negative.

Such erasure deletes the user's records but not the user's relationships, because every deleted rating had a counterparty whose other ratings stay public. Figure 1 shows a trader u who traded with users a, b and c. After erasure, these three users remain in the published graph with their other ratings; they were active in the same months as u, and, like most active traders, they are better connected than a typical user. An adversary who knows that u used the platform, such as a former counterparty, an investigator or a data broker, can therefore rank the remaining users by how likely each is to be one of u's *former neighbors*, the users that u rated or was rated by. Trading relationships are financial associations and trust relationships reveal social affiliations, so such a ranking is a privacy risk even when no single relationship is certain. The same risk arises for dataset custodians, who honor takedown requests by removing a user from a published graph [17, 21], and for machine-learning pipelines, whose exact unlearning retrains on the graph that remains [3, 4].

Existing tools leave this risk unmeasured. Link prediction [18] scores pairs of users who are both still in the graph, mostly through the neighbors they share. An erased user has no neighbors left, so every neighborhood-based score is zero, and preferential attachment, once the erased user's degree is set to a constant, ranks candidates by their own degree. Missing-node inference and cold-start link prediction [10, 16] reconstruct nodes or links that were never observed, whereas an erased user is known and was removed deliberately. Graph

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unlearning [5, 7] removes a user's influence from a trained model, which at best makes the model equivalent to one trained on the remaining graph; what that graph itself reveals lies outside the guarantee of unlearning.

Even an attack that ranks former neighbors well would leave the risk unexplained, because a high accuracy need not reveal anything specific to the erased user. Former neighbors tend to be well connected, so ranking users by degree already finds many of them, as it would in any graph in which the same users have the same degrees. Likewise, users who were active in the same months as the erased user are plausible partners in any graph with the same activity over time. An accuracy figure therefore cannot tell a platform whether its published graph exposes the erased user's relationships or only the popularity and timing of its users, yet the answer decides which remedy can work. What is missing is a method that measures this residual inference under explicitly stated adversary knowledge and attributes it to degree, to activity timing, and to what remains beyond them.

This paper asks how much the published graph reveals about an erased user's former neighbors, and why. We propose ChronoTrace, a benchmark that pairs an attack with a calibration. The attack ranks every remaining user under two separate levels of knowledge: a *release-only* adversary sees the published graph and the erased user's pseudonym, and a *window-informed* adversary also knows when the erased user was active, exactly or approximately, from side information such as old public posts or receipts of past trades. Either adversary trains on erasures that it simulates on the published graph itself, so it needs no ground truth about the erased user and never sees the user's own ratings. The calibration runs the same adversary on *reference graphs* that keep every user's degree, or every user's positive and negative activity in each period, and randomize who interacted with whom; two degree controls measure the accuracy that remains among candidates of equal degree. The gap between the real graph and each reference attributes the adversary's accuracy to degree, to activity timing, or to what remains beyond them. Unlike link prediction, ChronoTrace needs no neighborhood of the erased user; unlike graph unlearning, it examines the published graph rather than a model; and unlike an accuracy figure, it says where the accuracy comes from.

We evaluate ChronoTrace on three signed trust graphs with genuine timestamps: BitcoinOTC, BitcoinAlpha and dated Epinions. Our contributions are as follows.

- **An erasure attack with explicit adversary knowledge** (Sections 3 and 4). Without timing knowledge it ranks former neighbors with AUC 0.87–0.93, the probability that a former neighbor is ranked above a non-neighbor (0.5 is chance); even with the exact window, the share of former neighbors among its top five candidates is at most 0.19.
- **A calibration that attributes the inference** (Section 4.4). Degree explains all inference without a time window: among candidates of equal degree the adversary performs at chance (degree-matched AUC 0.46–0.52). On the two Bitcoin trading graphs, the activity window adds signal beyond degree (degree-matched AUC 0.81) and ranks low-degree former neighbors above chance, as long as it covers the user's active months; on Epinions, whose trust statements are seldom reciprocated, the window does not improve the ranking.
- **Two practical side channels** (Sections 6.1 and 6.3). An account directory that keeps users without remaining ratings visible identifies those former neighbors exactly. In the BitcoinOTC release, user IDs follow the order of first activity and let a release-only adversary recover about half of the exact window's gain, +0.018 AUC over ranking by degree.

The published graph also indicates the sign and the approximate time of a deleted relationship, and the two simple defenses we test, local edge suppression and edge-level randomized response, leave the degree and timing signals largely intact. We release the code and every per-target result, from which our scripts regenerate every number, table and figure. Section 2 positions the work, Sections 3 to 5 define the problem, the benchmark and the setup, Section 6 answers five research questions, and Sections 7 and 8 discuss implications and limitations.

2 Background and Related Work

Trust graphs and erasure in practice. A trust graph records ratings between users: who rated whom, when, and whether the rating was positive or negative. Reputation systems compute trust scores from such graphs to warn users about fraudulent counterparties [15], and review sites use them to filter content and recommendations [19]. Both endpoints of every edge are people, which distinguishes these graphs from bipartite user–item rating graphs, where one side consists of products and the relationship between two people is never recorded; we therefore study user–user graphs. Erasure from a user–item graph raises the analogous question for a user's purchases or reviews; the benchmark applies to it without change, but we do not evaluate it. The BitcoinOTC, BitcoinAlpha and Epinions graphs are public research releases [17, 21]. When a user leaves, a platform or custodian typically deletes the user's node with every incident edge, which is the deletion operator we study.

Link prediction, cold starts and missing nodes. Link prediction ranks node pairs by proximity in an observed graph, using common neighbors, Jaccard similarity, Adamic–Adar, Katz or preferential attachment [1, 18]. These scores need the neighborhoods of both endpoints, which an erased user lacks, so after erasure they reduce to zero or to a ranking by degree (Section 1). Section 6.2 shows through calibration that features which need no time window add nothing to this ranking; what lies beyond it comes from timing knowledge and from side channels. Cold-start link prediction predicts links for nodes without network data from side information

such as group memberships [16]. Missing-node identification and network completion infer nodes and edges absent from an observed graph [10, 14]. Our setting differs from these in three ways: the deleted user is known and its absence is deliberate; the question is which retained users were its neighbors; and every accuracy is calibrated against reference graphs that preserve degree or activity, which tells a platform whether the published graph reveals anything beyond those properties.

Privacy attacks on graph models and graph unlearning. LinkTeller [28], GNNBleed [23] and black-box edge attacks [12] recover edges from a trained graph neural network, and further work infers node membership [13], subgraph structure [27] or graph properties [31] from trained models. Graph unlearning removes deleted nodes or edges from trained models, by sharded retraining [5], with certified guarantees [7], through influence functions [29], in inductive settings [26], or through layer-wise deletion operators [6, 24]; Said et al. survey the area [22]. All of these treat a trained model as the attack surface or the object of removal. We study the published graph itself, which bounds what any model trained on it after exact unlearning can still encode.

De-anonymization. Structural re-identification links anonymized graph releases to identities through auxiliary graphs or planted structures [2, 11, 20]. Our adversary already knows the erased user's identity and asks which retained users were its neighbors. Table 1 summarizes this positioning.

3 Problem Statement and Formalism

3.1 Data model

A trust graph $G = (V, E)$ consists of users V and rating events E. Each event is

$$e = (s, d, \sigma, t), \tag{1}$$

where user s rated user d at time t with sign $\sigma \in \{-1, +1\}$; ratings above zero map to $+1$. In our datasets each ordered pair rates at most once and ratings are never withdrawn, so an edge is a single timestamped event, and no assumption about how long a relationship lasts is needed. Two users are *neighbors* if at least one event links them in either direction. A user's *degree* is its number of events, given or received, and its *neighbor count* $|N(v)|$ is its number of distinct neighbors. To summarize activity, we divide each dataset's time span into 32 equal *time bins*, about two months on the Bitcoin graphs and one month on Epinions; the bins serve only to compute features.

3.2 Erasure and the published graph

Erasing user u deletes every event in which u is the rater or the rated user,

$$E_u = \{(s, d, \sigma, t) \in E : \ s = u \ \vee \ d = u\}, \tag{2}$$

and the platform publishes

$$G_u^{\text{rel}} = (V \setminus \{u\}, \ E \setminus E_u), \tag{3}$$

which we call the *published graph*. Publication takes one of two forms. An *edge-list release*, the format of public research datasets, lists events only, so a user left without events disappears. A *directory release*, such as a live platform's list of accounts, still lists every user. The distinction matters: in a directory release any user left without events must have interacted only with u, so those former neighbors are identified exactly. Section 6.1 measures this channel; every other result uses edge-list releases, so those results measure inference beyond this lookup. The *candidates* C_u are the users present in the published graph.

3.3 Sensitive target

The protected information is the set of u's *former neighbors*,

$$N_u = \{v \in V \setminus \{u\} : \ \exists (s, d, \sigma, t) \in E_u \text{ with } v \in \{s, d\}\}. \tag{4}$$

Membership in N_u ignores direction, sign and time. Section 6.4 adds two secondary targets for each identified former neighbor: whether the deleted relationship included a negative rating, and when it happened.

3.4 Attack task and threat model

The adversary knows u's pseudonym, observes G_u^{rel}, and may hold side information A_u about when u was active. It outputs a score

$$\text{score}(G_u^{\text{rel}}, u, v, A_u) \quad \text{for every } v \in C_u, \tag{5}$$

and ranks the candidates by it. We separate two levels of knowledge, and every result states which level produced it.

- A *release-only* adversary has $A_u = \emptyset$: it reads only G_u^{rel} and u's pseudonym. It can rank candidates by features that need no time window, or estimate u's activity window from G_u^{rel} (Section 4.3).
- A *window-informed* adversary also knows u's activity window $W_u = [b_0, b_1]$, the time bins spanned by the events in E_u, padded by one bin on each side. We evaluate the exact window, an upper bound on timing knowledge, and the exact window shifted by δ bins in a direction the adversary does not know.

Figure 1 instantiates these definitions: $N_u = \{a, b, c\}$, W_u is the shaded band, and the candidates are a to i. The threat model also excludes stronger adversaries. The adversary holds no copy of the graph from before the erasure, because comparing two snapshots would reveal N_u directly, and it attacks one erasure at a time. Recovering edge directions, attacking storage, and inverting a separately trained model are out of scope.

3.5 Metrics

A ranking is scored over all candidates. AUC, the area under the ROC curve, is the probability that a randomly chosen former neighbor ranks above a randomly chosen non-neighbor, so 0.5 is chance. AP (average precision) averages the precision at the rank of each former neighbor. P@k is the fraction of the top k candidates that are former neighbors, R@k the fraction of former neighbors in the top k, and

Table 1: Positioning against related work.

Work	Infers	Attack surface	Timestamps	Signs	Target node present	Calibration
Liben-Nowell and Kleinberg [18]	future links	observed graph	split only	no	yes	none
Leroy et al. [16]	links of a node without edges	side information	no	no	no edges	none
Eyal et al. [10], Kim and Leskovec [14]	missing nodes and edges	observed graph	no	no	unknown	none
Backstrom et al. [2], Hay et al. [11]	node identity	released graph	no	no	yes	none
Narayanan and Shmatikov [20]	node identity	auxiliary graph	no	no	yes	none
LinkTeller [28], Maui [12]	training edges	GNN queries	no	no	yes	none
GNNBleed [23]	private edges	GNN access	yes	no	yes	none
Zhang et al. [31]	properties, subgraphs	GNN access	no	no	yes	none
GraphEraser [5], Chien et al. [7]	(removes model influence)	model parameters	no	no	removed	–
This work	**former neighbors, sign, time**	**published graph**	**yes**	**yes**	**erased**	**degree, activity**

S@k is 1 if at least one former neighbor is in the top k; the best and mean ranks locate the former neighbors in the list. Two further metrics isolate what degree cannot explain. The *degree-matched AUC* compares only former neighbors and non-neighbors with the same number of events in G_u^{rel}, so ranking by degree scores exactly 0.5. The *excess over degree* is the AUC minus the AUC expected if the former neighbors were redrawn at random among candidates with the same degrees (Section 4.4).

3.6 Difficulty

Exact recovery by guessing is hopeless for most users. Between 64% and 79% of the users in our datasets have at most three neighbors, and guessing even a three-member N_u exactly at random among 5,880 BitcoinOTC candidates succeeds with probability below 10^{-10}. A ranking is therefore the meaningful output, and we report its quality at every depth.

4 The ChronoTrace Benchmark

Figure 2 shows the benchmark for one erasure. The platform deletes u's events, optionally applies a defense, and publishes the result. The adversary describes every candidate by features of the published graph, adds timing features if it holds a window, ranks the candidates with a model trained on erasures it simulates itself, and is scored against N_u. The same pipeline runs on reference graphs to calibrate the result.

4.1 What the published graph reveals

ChronoTrace describes each candidate v by eleven features computed on G_u^{rel} (Table 2; exact definitions in Appendix A). Popularity is captured by v's degree (feature 0). Signs and directions enter through v's balance of positive over negative events and of outgoing over incoming events (features 1 and 2). Local structure enters through the mean degree of v's neighbors, the number of users two hops away, and a closure ratio (features 7 to 9). Time enters in two ways. Feature 10 measures how concentrated v's activity is across the 32 bins and needs no knowledge of u. Features 3 to 6 align v's activity with the window W_u: the share of v's events inside the window, the change in v's activity across the window boundaries, and the shares of v's positive and of its negative events inside the window. Features 3 to 6 need a window: the exact or shifted window of a window-informed adversary, or a window that a release-only adversary estimates from G_u^{rel}.

4.2 Adversaries and how they train

Each adversary scores candidates with a logistic regression on standardized features, with balanced class weights. We use a linear model so that its coefficients show which evidence drives the ranking (Section 6.2); Appendix D.9 shows that a random forest ranks no better. We name an adversary by its feature set (Table 2) and, for sets that need a window, by its window. The *window-free* sets are Degree, Topology and Full-R, which uses every feature that needs no window. The *window-based* sets are Temporal, Sgn-Struct (signed-structural) and ChronoTrace, which uses all eleven features; we name this full attacker after the benchmark.

An adversary needs labeled examples, and it can make them from G_u^{rel} alone. To attack the erasure of u, it picks other users $u_1, \ldots, u_m$, erases each from G_u^{rel} itself, and labels each simulated erasure with the neighbors that u_j has in G_u^{rel}. These simulated *shadow erasures* contain no event of u and no label that mentions u, so nothing about the target reaches training. The adversary pools the shadow erasures, keeps every former neighbor and $\max(1000, 20 \times \text{positives})$ non-neighbors, and fits the model. Any users present in G_u^{rel} can serve as shadow users; we use the other eleven targets drawn with the same seed, so that the two training protocols compared in Section 6.7 train on the same users.

4.3 Timing knowledge

The exact window W_u is the benchmark's upper bound on timing knowledge. Because W_u is padded by one bin, a window shifted by one bin still covers every bin in which u was active, and larger shifts miss some of them. To model imperfect knowledge, we shift W_u by $\delta \in \{1, 2, 4, 8\}$ bins in a random direction, keeping its width, and shift the window of every shadow erasure by the same δ in an independently drawn direction, so that training sees errors of the same size as the test. Two estimators use only G_u^{rel} and u's pseudonym. The *hub* estimator centers a seven-bin window on the median time of the published events of the 50 users with the most neighbors; it depends on the target only through the deleted events and in our data gives every target of a dataset the same window. The *ID-cohort* estimator exploits pseudonyms numbered in order of first activity: it takes the 25 users whose IDs are closest to u's and returns the median of their first active bins and of their last active bins, padded by one bin. We report the intersection over union (IoU) of every estimate with W_u.

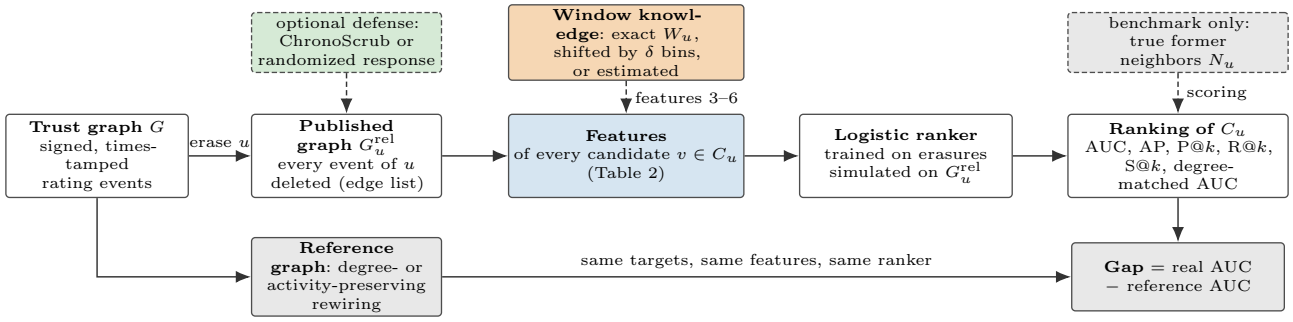


Figure 2: The ChronoTrace benchmark for one erasure. The window is exact, shifted or estimated from the published graph; the adversary trains only on erasures it simulates on that graph.

Table 2: Candidate features, the information each reads, and the feature sets that use it.

			Window-free			Window-based		
#	Feature	Reads	Degree	Topology	Full-R	Temporal	Sgn-Struct	ChronoTrace
0	Degree	degree	✓	✓	✓			✓
1	Sign balance	sign			✓		✓	✓
2	Direction balance	direction			✓		✓	✓
3	Window activity share	time, W_u				✓		✓
4	Activity shift at window	time, W_u				✓		✓
5	Positive window share	time, sign, W_u					✓	✓
6	Negative window share	time, sign, W_u					✓	✓
7	Neighbor degree	structure		✓	✓		✓	✓
8	Two-hop reach	structure		✓	✓		✓	✓
9	Closure ratio	structure		✓	✓		✓	✓
10	Activity concentration	time			✓	✓		✓

4.4 Calibration

An AUC alone measures risk but not its cause, and the cause decides which defense can work. ChronoTrace therefore compares each adversary's AUC on the real graph with its AUC on *reference graphs* that keep some properties of the real graph and randomize the rest.

- The *degree reference* rewires the graph by permuting the rated end of every event at random. Every user keeps its number of events given and received, and the partners become random. Its *label reference* draws $|N_u|$ pseudo-neighbors from the real candidates with probability proportional to degree, which approximates u's neighbors under the rewiring.
- The *activity reference* permutes the rated end only among events in the same time bin with the same sign. Every user then keeps its number of positive and of negative events given and received in every bin, so its degree and every timing feature are unchanged, and only who interacted with whom is randomized.

We erase the same targets in each reference graph and run the same adversary. The *gap* to a reference, the real AUC minus the reference AUC, measures how much better the adversary ranks the true former neighbors than neighbors that are random given the preserved properties. The deletion is the same in both graphs, so a gap reflects how partners were chosen in the original graph beyond the preserved properties. It can come from the reciprocity of ratings, from community structure or from how our targets were selected, as well as from the erased user's own relationships.

Two *degree controls* complement the rewiring. The degree-matched AUC restricts the comparison to candidates of equal degree. The *excess over degree* is the AUC minus the AUC expected when the former neighbors are redrawn at random within each degree class, which keeps their exact degree histogram. By linearity of expectation, the expected AUC under this redraw equals $(\sum_b p_b \bar{r}_b - |N_u|(|N_u|+1)/2)/(|N_u||C_u \setminus N_u|)$, where p_b counts former neighbors of degree b and $\bar{r}_b$ is the mean rank of degree-b candidates; in this formula only, ranks count upward from 1 for the lowest score.

4.5 Adapted heuristic baselines

Classical link-prediction scores need both endpoints, and the erased user has no edges left. We therefore adapt them into scores of each candidate against an *anchor set*: the candidates most active inside the exact window, with $\max(5, \min(50, 3|N_u|))$ anchors. The anchor set reads the exact window and $|N_u|$, so every anchor-based heuristic is optimistic: it uses knowledge that a realistic attack lacks. Common neighbors, Jaccard and Adamic–Adar compare a candidate's neighborhood with the anchor set; the *anchor overlap* of a candidate, its number of neighbors among the anchors, is its common-neighbors score. Preferential

attachment multiplies the candidate's neighbor count by the anchors' mean neighbor count. The latter is constant within a target, so preferential attachment ranks exactly like the neighbor-count prior $\log(1 + |N(v)|)$. The window activity share and activity shift heuristics are features 3 and 4. Signed motif adds the absolute sign balance, the positive and negative window shares, and $\log(1 + \text{anchor overlap})$. *CT-Unsupervised* is a fixed, untrained combination of these signals: $\log(1 + |N(v)|) + \log(1 + \text{anchor overlap}) + 2 \times$ window activity share + activity shift + signed motif.

4.6 Defenses

We evaluate two defenses a platform could apply before publishing. ChronoScrub is a diagnostic edge suppression that uses what the platform knows at deletion time, N_u and W_u. It marks the former neighbors and their neighbors as residue users. Its temporal variant deletes 45% of the events inside the window that touch a residue user, chosen by a deterministic hash of each event. Its joint variant at strength $s \in \{0.25, 0.50, 0.75\}$ deletes a share $0.45s$ of those events and also a share $0.25s$ of the events between two residue users. ChronoScrub offers no formal guarantee. Edge-level randomized response publishes each remaining event independently with probability $e^{\epsilon}/(1 + e^{\epsilon})$ for $\epsilon \in \{0.5, 1, 2, 4\}$; it only deletes events, so it falls short of edge-level differential privacy. The adversary knows the defense and simulates its erasures on the defended release; under ChronoScrub it also applies ChronoScrub to each simulated erasure, as the platform would. We also measure utility as the AUC of predicting the sign of the latest 20% of events from the defended release (Appendix B.6).

5 Experimental Setup

Datasets. We use three signed trust graphs with genuine timestamps (Table 3). BitcoinOTC and BitcoinAlpha record trader ratings on two over-the-counter bitcoin markets [15, 17]. Dated Epinions consists of the trust and distrust statements of the Extended Epinions release that carry a genuine creation date [19, 21]. The first recorded date of that release is a placeholder shared by 68.8% of all statements, and we drop those statements together with self-ratings. We number the dated users in order of their first rating, so the ID–time correlation of dated Epinions in Table 3 reflects this numbering rather than the platform's. The widely used static SNAP Epinions snapshot has no timestamps, only record order; Appendix E evaluates it as well.

Targets and protocol. For each of five seeds we draw 12 target users per dataset, stratified over the quartiles of incident-event count between 4 and 128, which gives 60 erasures per dataset. The targets have between 2 and 99 distinct former neighbors (median 6), fewer than their events because many ratings are reciprocated. Section 6.4 adds 60 targets per dataset with one to three events, a group that contains 45% to 78% of all users. Every erasure is evaluated on its full candidate universe, without subsampling. Intervals are percentile-bootstrap 95% confidence intervals over targets (10,000 resamples), differences are paired by target, and p-values come from two-sided sign-flip permutation tests with 10,000 random flips. Appendix B details the protocol. Comparisons between adversaries are Benjamini–Hochberg corrected within each dataset (Appendix D.2).

6 Results

We organize the results around five research questions, RQ1 to RQ5, each ending with an explicit answer; two subsections on heuristic baselines and validity checks follow.

6.1 RQ1: How much does the published graph reveal?

Table 4 reports every adversary, and Figure 3 shows the same AUCs with their confidence intervals. The Degree adversary reaches AUC 0.910 on BitcoinOTC, 0.869 on BitcoinAlpha and 0.924 on Epinions. Adding signs, directions, structure and activity concentration does not improve on degree alone: neither Topology nor Full-R is significantly better than Degree on any dataset, and Full-R is slightly worse on BitcoinOTC (0.901; Appendix D.2). Figure 4 shows the per-target differences behind these comparisons; for both window-free sets they scatter around zero.

The ranking is useful for narrowing a search but rarely pinpoints a neighbor. With degree alone, the share of former neighbors among the top five candidates is 0.13, 0.08 and 0.10, and a former neighbor appears in the top ten for 68%, 50% and 35% of targets. Appendix D.1 reports precision, success and ranks at further depths.

A directory release adds an exact channel. In such a release a user left without events must have interacted only with the erased user. Such isolated users make up 2.3%, 3.9% and 6.6% of all former neighbors, and at least one appears for 15%, 25% and 48% of targets. Ranking by degree places them last, so a directory release also lowers the Degree adversary's AUC to 0.895, 0.838 and 0.830. Appendix E reports every adversary on directory releases.

Answer. The published graph lets an adversary rank former neighbors far above chance, with low top-of-list precision. Keeping isolated accounts visible discloses some former neighbors exactly. In practice, erasure does not hide a user's partners from someone who knows the user existed: on BitcoinOTC the 500 best-ranked of 5,880 candidates contain 75% of former neighbors.

6.2 RQ2: What explains inference without a window?

Degree explains the window-free adversaries completely. Even with every window-free feature (Full-R), the degree-matched AUC is 0.51, 0.52 and 0.46, and every confidence interval contains 0.5 (Table 5). Full-R's excess over degree is -0.001, $+0.006$ and $+0.002$, none significant. Topology's degree-matched AUC is 0.52 [0.45, 0.58], 0.53 [0.47, 0.59] and 0.55 [0.49, 0.61], so its intervals contain 0.5 as well. Window-free features thus add nothing to degree.

Table 3: Datasets. Bin: days per time bin. Distrust: share of negative ratings. Recip.: share of ratings whose reverse exists; ≤7d: reversed within 7 days. ID ρ: Spearman correlation of user ID and first activity. ≤3 nbrs: users with at most three neighbors. Het.: degree heterogeneity $\overline{d^2}/\bar{d}^2$.

Dataset	Users	Ratings	Period	Bin	Distrust	Recip.	≤7d	ID ρ	≤3 nbrs	Het.
BitcoinOTC	5,881	35,592	2010-11 to 2016-01	59	10%	0.79	68%	1.00	65%	11
BitcoinAlpha	3,783	24,186	2010-11 to 2016-01	59	6%	0.83	70%	0.33	64%	8
Epinions	56,499	261,803	2001-01 to 2003-08	29	20%	0.29	18%	0.59	79%	30

Table 4: Former-neighbor ranking on edge-list releases (mean over 60 targets; AUC with 95% CI). Random: random scores. Window: none, estimated from the published graph, or exact. Features: indices from Table 2. DM-AUC: degree-matched AUC. Bold: highest AUC.

Dataset	Feature set	Window	Features	AUC [95% CI]	AP	P@5	S@10	R@100	R@500	DM-AUC
BitcoinOTC	Random	–	–	0.495 [0.465, 0.525]	0.002	0.00	0.02	0.006	0.075	–
	Degree	none	0	0.910 [0.889, 0.930]	0.111	0.13	0.68	0.447	0.751	0.50
	Topology	none	0, 7–9	0.908 [0.886, 0.928]	0.116	0.11	0.67	0.456	0.741	0.52
	Full-R	none	0–2, 7–10	0.901 [0.877, 0.922]	0.089	0.09	0.58	0.424	0.734	0.51
	ChronoTrace	hub (from release)	0–10	0.899 [0.875, 0.921]	0.073	0.08	0.52	0.406	0.730	0.55
	ChronoTrace	ID cohort (from release)	0–10	0.928 [0.911, 0.944]	0.112	0.11	0.62	0.516	0.777	0.70
	Temporal	exact	3, 4, 10	0.939 [0.927, 0.951]	0.101	0.13	0.63	0.490	0.772	0.78
	Sgn-Struct	exact	1, 2, 5–9	0.946 [0.933, 0.957]	0.145	0.16	0.73	0.536	0.825	0.80
	ChronoTrace	exact	0–10	**0.949** [0.937, 0.960]	0.145	0.19	0.72	0.575	0.826	0.81
BitcoinAlpha	Random	–	–	0.474 [0.436, 0.512]	0.004	0.00	0.03	0.016	0.119	–
	Degree	none	0	0.869 [0.838, 0.898]	0.087	0.08	0.50	0.369	0.718	0.50
	Topology	none	0, 7–9	0.871 [0.842, 0.898]	0.071	0.07	0.48	0.358	0.715	0.53
	Full-R	none	0–2, 7–10	0.868 [0.838, 0.895]	0.067	0.07	0.40	0.347	0.711	0.52
	ChronoTrace	hub (from release)	0–10	0.856 [0.822, 0.888]	0.061	0.05	0.35	0.334	0.702	0.61
	ChronoTrace	ID cohort (from release)	0–10	0.880 [0.847, 0.909]	0.070	0.07	0.45	0.374	0.726	0.57
	Temporal	exact	3, 4, 10	0.907 [0.885, 0.928]	0.092	0.09	0.53	0.419	0.801	0.80
	Sgn-Struct	exact	1, 2, 5–9	0.930 [0.912, 0.947]	0.127	0.11	0.60	0.534	0.843	0.81
	ChronoTrace	exact	0–10	**0.931** [0.914, 0.947]	0.103	0.11	0.58	0.554	0.837	0.81
Epinions	Random	–	–	0.479 [0.451, 0.508]	0.000	0.00	0.00	0.000	0.004	–
	Degree	none	0	0.924 [0.903, 0.944]	0.051	0.10	0.35	0.189	0.401	0.50
	Topology	none	0, 7–9	0.926 [0.904, 0.946]	0.034	0.05	0.23	0.173	0.409	0.55
	Full-R	none	0–2, 7–10	0.922 [0.900, 0.942]	0.034	0.05	0.27	0.157	0.413	0.46
	ChronoTrace	hub (from release)	0–10	0.919 [0.898, 0.939]	0.029	0.04	0.27	0.164	0.390	0.48
	ChronoTrace	ID cohort (from release)	0–10	0.926 [0.905, 0.945]	0.028	0.04	0.22	0.155	0.399	0.59
	Temporal	exact	3, 4, 10	0.903 [0.879, 0.927]	0.022	0.04	0.27	0.128	0.342	0.61
	Sgn-Struct	exact	1, 2, 5–9	**0.926** [0.905, 0.946]	0.051	0.08	0.35	0.209	0.466	0.60
	ChronoTrace	exact	0–10	0.923 [0.899, 0.944]	0.030	0.04	0.22	0.173	0.428	0.59

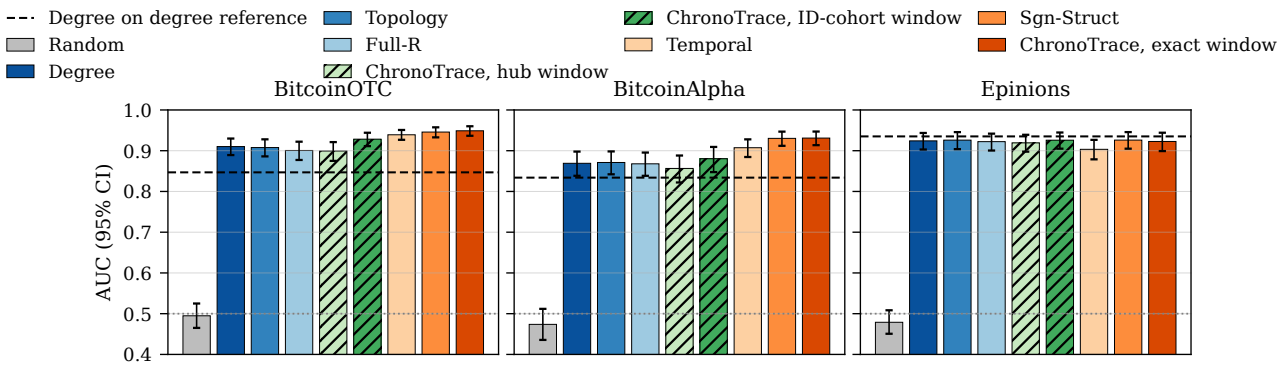


Figure 3: AUC of every adversary (95% CI over 60 targets). Blue: no window; green: window estimated from the published graph; orange: exact window. Dashed: Degree on the degree reference; dotted: chance.

The degree reference shows how much of the Degree adversary's advantage over chance any graph with the same degrees would give (Figure 5). On the rewired graphs, ranking by degree reaches 0.847, 0.834 and 0.935, against 0.910, 0.869 and 0.924 on the real graphs, so most of the advantage

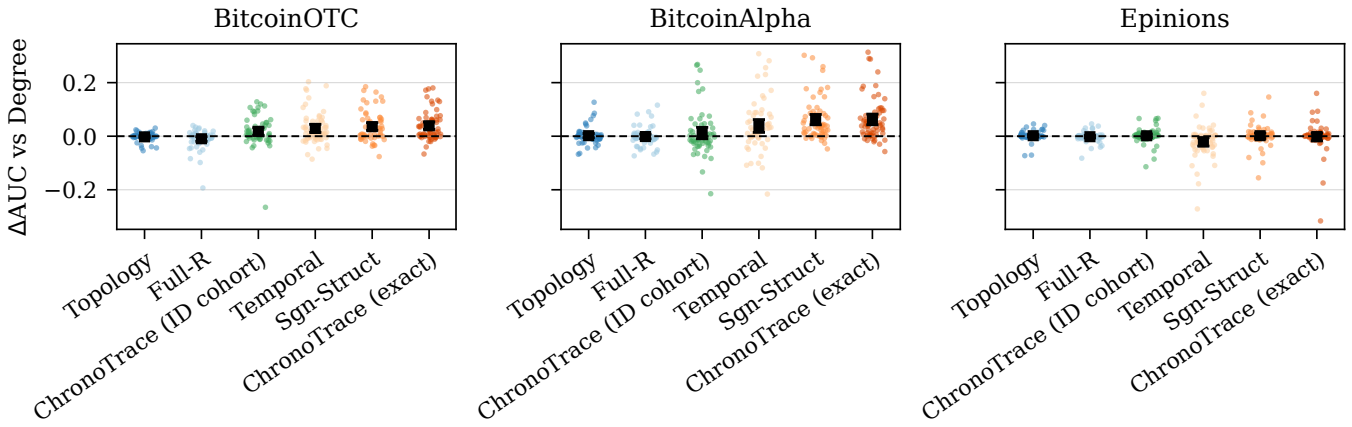


Figure 4: Per-target AUC difference to the Degree adversary (points), with mean and 95% CI (black).

Table 5: AUC on the real graph and on reference graphs (rewired: mean over independent rewirings), with degree controls; 60 targets per dataset. Degree (labels): label reference. DM-AUC: degree-matched AUC.

			Reference-graph AUC				
Dataset	Feature set	Real AUC	Degree (labels)	Degree (rewired)	Activity (rewired)	DM-AUC [95% CI]	Excess over degree
BitcoinOTC	Degree	0.910	0.856	0.847	0.844	0.50 [0.50, 0.50]	0.000
	Full-R	0.901	–	0.847	0.849	0.51 [0.45, 0.59]	−0.001
	ChronoTrace (exact)	0.949	–	0.843	0.932	0.81 [0.76, 0.86]	+0.079*
BitcoinAlpha	Degree	0.869	0.852	0.834	0.829	0.50 [0.50, 0.50]	0.000
	Full-R	0.868	–	0.837	0.828	0.52 [0.46, 0.58]	+0.006
	ChronoTrace (exact)	0.931	–	0.836	0.914	0.81 [0.77, 0.85]	+0.109*
Epinions	Degree	0.924	0.913	0.935	0.943	0.50 [0.50, 0.50]	0.000
	Full-R	0.922	–	0.935	0.940	0.46 [0.39, 0.52]	+0.002
	ChronoTrace (exact)	0.923	–	0.932	0.948	0.59 [0.52, 0.65]	+0.011

* $p < 0.05$, two-sided sign-flip permutation test (unadjusted).

over chance is present in any graph with the same degrees. On the Bitcoin graphs the gaps that remain, +0.063 (95% CI [+0.042, +0.083]) and +0.035 (95% CI [+0.001, +0.068]), mean that the erased users' partners are somewhat better connected than random partners of users with the same degrees; on Epinions the gap of −0.011 (95% CI [−0.032, +0.008]) is not significant. As Section 4.4 explains, these gaps describe how partners were chosen in the original graph rather than anything the deletion creates. On directory releases the same gap can turn negative, because ranking by degree places the isolated former neighbors of Section 6.1 last (Appendix E).

The model coefficients point the same way (Table 6). In the Full-R model the largest weights fall on degree or on two-hop reach, which grows with degree. Neighbor degree carries a negative weight in every model, so the model prefers well-connected users whose own neighbors are not hubs. Because the features are correlated, the weights show the direction of each feature's influence rather than its unique contribution, and the AUC confirms that together they add nothing to degree.

Ranking by degree has a blind spot: it misses low-degree former neighbors (Table 7; Appendix D.7). It places former neighbors with one to three events at AUC 0.30, 0.28 and 0.49, at or below chance, and those with more than 20 events at 0.96 or above. Figure 1(c) illustrates the risk: a hub such as d, never linked to u, can outrank true former neighbors. Without a window, the attack therefore exposes mainly a user's relationships with well-connected users.

Table 6: Mean standardized logistic coefficients over 60 targets (OTC: BitcoinOTC; Alpha: BitcoinAlpha; Epin.: Epinions).

		Full-R			ChronoTrace (exact)		
#	Feature	OTC	Alpha	Epin.	OTC	Alpha	Epin.
0	Degree	+1.33	+0.89	+1.01	+1.21	+0.49	+0.93
1	Sign balance	−0.11	+0.18	+0.14	+0.11	+0.50	+0.12
2	Direction balance	+0.30	+0.14	+0.05	+0.27	+0.13	+0.06
3	Window activity share				+1.31	+0.71	+0.37
4	Activity shift at window				+0.01	+0.17	+0.04
5	Positive window share				+0.46	+1.07	+0.52
6	Negative window share				+0.14	+0.24	+0.13
7	Neighbor degree	−0.39	−0.87	−1.23	−0.31	−0.98	−1.03
8	Two-hop reach	+0.22	+0.86	+1.70	+0.10	+1.02	+1.43
9	Closure ratio	−0.05	−0.06	−0.12	−0.06	−0.05	−0.08
10	Activity concentration	−0.14	−0.01	−0.15	−0.59	−0.38	−0.29

Answer. On the three dated graphs, an adversary without a window learns nothing about the erased user beyond

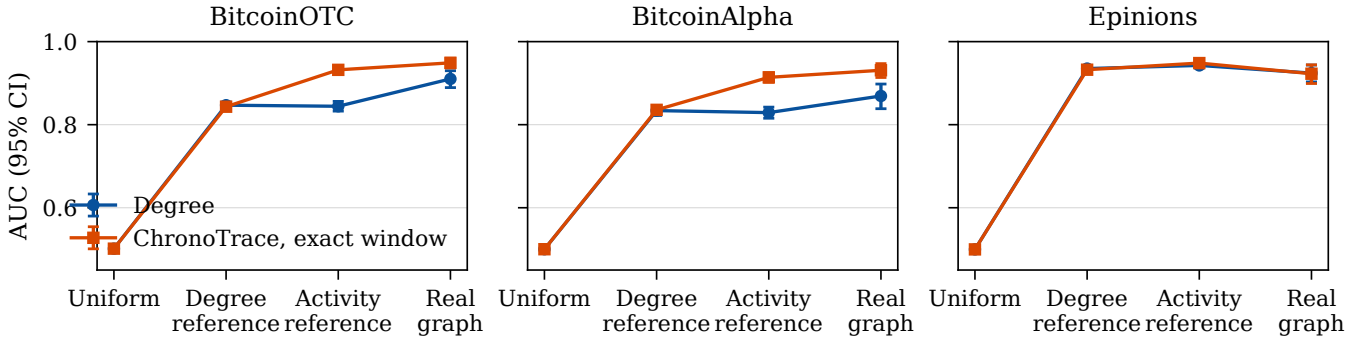


Figure 5: AUC of the Degree and ChronoTrace (exact window) adversaries when the former neighbors are drawn uniformly at random, drawn at random given degrees or given every user's activity, and real (95% CI).

Table 7: AUC of former neighbors grouped by their degree in G_u^{rel}, each group against all non-neighbors.

	BitcoinOTC			BitcoinAlpha			Epinions		
Degree in G_u^{rel}	1–3	4–20	>20	1–3	4–20	>20	1–3	4–20	>20
Degree	0.30	0.76	0.98	0.28	0.74	0.96	0.49	0.89	0.99
Full-R	0.39	0.74	0.97	0.32	0.73	0.96	0.49	0.88	0.99
ChronoTrace (exact)	0.69	0.87	0.98	0.71	0.84	0.97	0.50	0.88	0.99
Former neighbors	23	97	423	26	108	309	64	137	767

the degree of each candidate, and the degree reference reproduces most of its advantage over chance. The risk falls on relationships with well-connected users.

6.3 RQ3: How much does knowing when help, and how accurately?

Exact timing adds signal that degree cannot explain. With the exact window, ChronoTrace reaches AUC 0.949 on BitcoinOTC and 0.931 on BitcoinAlpha, gains of +0.039 and +0.062 over Degree (Table 4). Its degree-matched AUC is 0.81 and 0.81, and its excess over degree is +0.079 and +0.109 (both $p < 0.001$). It also ranks low-degree former neighbors, which degree misses, above chance, at AUC 0.69 (95% CI [0.63, 0.76]) and 0.71 (95% CI [0.64, 0.78]). The window features carry large weights: the share of a candidate's events inside the window is the largest coefficient on BitcoinOTC (+1.31), and the share of its positive events inside the window is the largest on BitcoinAlpha (+1.07).

The reference graphs show where this signal comes from (Figure 5, Table 5). On the degree reference, ChronoTrace reaches only 0.843 and 0.836. On the activity reference, which keeps every user's per-bin activity but randomizes partners, it reaches 0.932 and 0.914. The activity reference thus reproduces 84% and 82% of ChronoTrace's advantage over the degree reference: most of the timing signal comes from users being active in the same months. The gap to the activity reference, +0.017 (95% CI [+0.004, +0.030]) and +0.017 (95% CI [+0.001, +0.031]), is accuracy that per-bin activity does not explain. Like the gap to the degree reference, it can reflect reciprocity, community structure or target

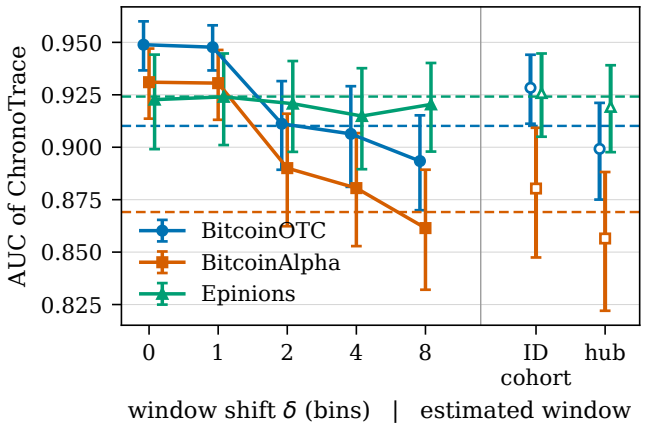


Figure 6: AUC of ChronoTrace as its window is shifted by δ bins (left) or estimated from the published graph (right). Dashed: Degree adversary.

selection as well as the erased user's own relationships. The trading graphs record shared activity as reciprocal ratings: 68% of BitcoinOTC ratings are reciprocated within a week. By contrast, on Epinions the window does not improve the ranking (0.923 against 0.924 for Degree), its signal among candidates of equal degree is weak (degree-matched AUC 0.59), and both reference graphs score at least as high as the real one. Epinions trust statements are reciprocated far less often (Table 3).

The gain requires a window that covers the user's active months (Figure 6; Appendix D.4). A window shifted by one bin, about two months, keeps the full gain, because with the one-bin padding of W_u it still covers every bin in which u was active. A shift of two bins, which misses part of that period, removes all of the gain on BitcoinOTC (+0.001 over Degree, not significant) and two thirds of it on BitcoinAlpha (+0.021, 95% CI [+0.009, +0.035]). A shift of eight bins removes it on both and leaves the BitcoinOTC adversary below Degree (−0.017).

A release-only adversary can instead estimate the window from the published graph. The hub estimator, which gives every target the same window (IoU 0.26 on BitcoinOTC), does not beat Degree. By contrast, the ID-cohort estimator succeeds where IDs follow the order of first activity. On BitcoinOTC, where ID and first activity have Spearman correlation 0.998, its windows reach IoU 0.66. They raise AUC over Degree by +0.018 (95% CI [+0.003, +0.031]), about half of the exact-window gain, using only the published graph and the erased user's pseudonym. On BitcoinAlpha, where the correlation is 0.334, the ID-cohort window still carries signal beyond degree (excess over degree +0.040, $p < 0.001$) but does not significantly raise AUC over Degree (+0.011). On dated Epinions, whose IDs we assigned in order of first rating, its excess over degree is small (+0.011). ID order is thus the one route beyond degree that we found for a release-only adversary.

Answer. Timing knowledge reaches relationships that degree cannot, including those of low-degree users, as long as the window covers the user's active months, and mainly on graphs whose edges record joint activity. Sequential pseudonyms supply enough timing on BitcoinOTC to recover about half of the gain without any side information, so whoever numbers the users of a release, platform or custodian, should not number them in order of first activity.

6.4 RQ4: Who is exposed, and what else is revealed?

On the trading graphs, users with few relationships are ranked no less accurately by AUC (Figure 7). When the 60 targets are split into three groups of about 20 by the number of their former neighbors that remain in the release, ChronoTrace reaches AUC 0.955 for the smallest neighborhoods on BitcoinOTC and 0.956 on BitcoinAlpha, no lower than for the largest (0.937 and 0.916). By contrast, on Epinions, where timing does not help, the smallest neighborhoods are the least exposed (0.898 against 0.951). For targets with only one to three events, ChronoTrace reaches 0.978 and 0.956 on the Bitcoin graphs, against 0.953 and 0.893 for Degree, although their former neighbors reach the top ten less often than those of the main targets (Appendix D.5).

The sign of a deleted relationship is also inferable. For every former neighbor present in G_u^{rel}, we use its remaining sign balance (feature 1) to predict whether its relationship with u included a negative rating. The prediction reaches AUC 0.82, 0.82 and 0.86; distrust occurs in 14%, 5% and 23% of these relationships. Users with many negative ratings are likely to have exchanged one with u too, so an adversary who finds a former neighbor can often also tell whether the two distrusted each other.

The time of a deleted relationship is inferable as well. For each identified former neighbor, we take the time bin with the most events in its remaining record and compare it with the bin of its earliest deleted event with u. The two coincide for 43% of former neighbors on BitcoinOTC, 47% on BitcoinAlpha and 31% on Epinions, against 9%, 26% and 4% when always guessing the dataset's busiest bin, and they lie within one bin of each other for 61%, 69% and 47%. A window-informed adversary that takes the neighbor's busiest bin inside W_u comes within one bin for 75%, 80% and 62% of them, against 56%, 50% and 31% for the middle bin of W_u (Appendix D.6).

Answer. On the trading graphs, AUC does not fall with the size of a user's neighborhood, and the published graph indicates the sign and the approximate time of a deleted relationship as well as its likely existence. Erasure requests from users with few relationships therefore deserve the same protection as those from active users.

6.5 RQ5: Do simple defenses help?

Neither defense of Section 4.6 changes the picture (Table 8, Figure 8). ChronoScrub removes up to 14% of the events. Because it deletes events of the former neighbors, it lowers the Degree adversary's AUC by up to 0.044, yet ChronoTrace loses at most 0.016. Randomized response at $\epsilon = 0.5$ removes 38% of all events yet leaves the Degree adversary at 0.904 on BitcoinOTC (0.910 without it), because random subsampling preserves which users are well connected. Utility barely changes under either defense; on BitcoinAlpha, sign prediction is weak to begin with. We expect that a defense would need to flatten the degree heterogeneity of the published graph, for example through degree-sequence anonymization or differentially private graph synthesis [30]; we did not evaluate such defenses.

Answer. Local suppression and edge-level randomized response leave the degree signal and the timing signal largely intact. A platform therefore cannot substantially reduce the risk by deleting more edges around the erased user; protection at the level of the published graph would likely require perturbing degrees themselves.

6.6 Adapted heuristic baselines

The adapted heuristics of Section 4.5 confirm these findings without training (Table 9; Appendix D.8). The neighbor-count prior, the only window-free heuristic besides the absolute sign balance, ranks close to the Degree adversary, and preferential attachment ranks identically to it. The anchor-based heuristics, which read the exact window and $|N_u|$, do not beat the trained ChronoTrace adversary; CT-Unsupervised comes closest on the Bitcoin graphs. The absolute sign balance ranks former neighbors below chance, mostly because users with one or two ratings have the largest absolute balance and are rarely former neighbors.

6.7 Validity checks

Four checks address how the adversary and the evaluation were set up. First, training on erasures simulated on the original graph, whose training graphs still contain u, gives the same AUC as training on the published graph to within 0.0008 for every feature set on every dataset, in both release formats (Appendix C.2). Second, every result above uses the full candidate universe. On Epinions, AUC is

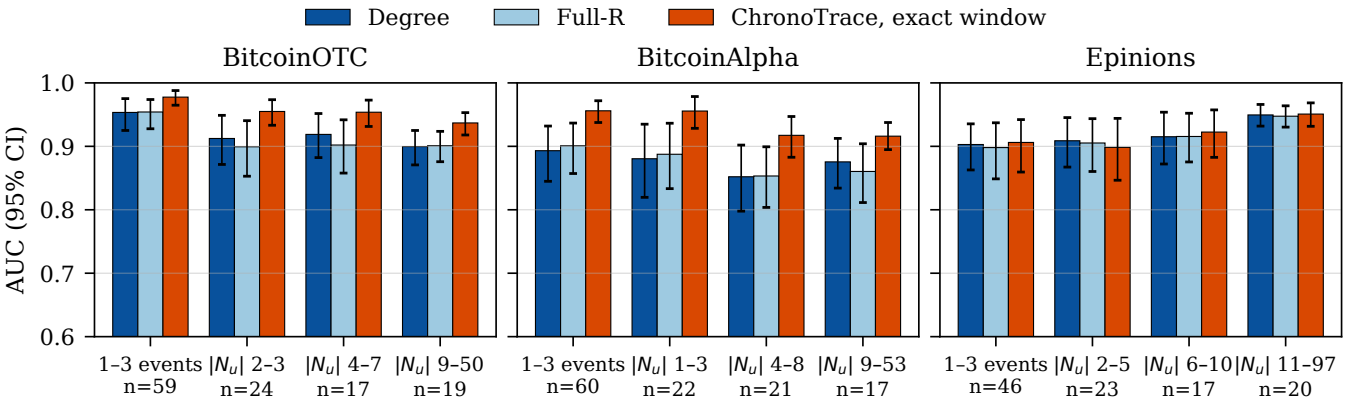


Figure 7: AUC by target group (95% CI): targets with one to three events, and the main targets split by their number of former neighbors $|N_u|$ that remain in the release.

Table 8: Defenses: mean AUC of the Degree and ChronoTrace (exact window) adversaries over 60 targets. Kept: share of events published. Utility: sign-prediction AUC on the defended release. RR: randomized response.

	BitcoinOTC				BitcoinAlpha				Epinions			
Defense	Degree	ChronoTrace	Kept	Utility	Degree	ChronoTrace	Kept	Utility	Degree	ChronoTrace	Kept	Utility
None	0.910	0.949	100%	0.635	0.869	0.931	100%	0.545	0.924	0.923	100%	0.767
ChronoScrub, temporal	0.875	0.933	91%	0.632	0.825	0.928	91%	0.544	0.920	0.918	86%	0.768
ChronoScrub, joint 0.25	0.901	0.943	97%	0.636	0.860	0.929	97%	0.545	0.928	0.926	96%	0.766
ChronoScrub, joint 0.50	0.895	0.942	94%	0.632	0.846	0.932	94%	0.549	0.928	0.926	92%	0.766
ChronoScrub, joint 0.75	0.886	0.939	92%	0.633	0.836	0.934	92%	0.550	0.920	0.921	88%	0.767
RR, $\epsilon = 0.5$	0.904	0.944	62%	0.639	0.856	0.925	62%	0.544	0.919	0.915	62%	0.767
RR, $\epsilon = 1$	0.907	0.946	73%	0.637	0.861	0.926	73%	0.544	0.927	0.924	73%	0.768
RR, $\epsilon = 2$	0.910	0.948	88%	0.637	0.867	0.930	88%	0.547	0.924	0.925	88%	0.764
RR, $\epsilon = 4$	0.910	0.949	98%	0.639	0.869	0.930	98%	0.548	0.923	0.922	98%	0.769

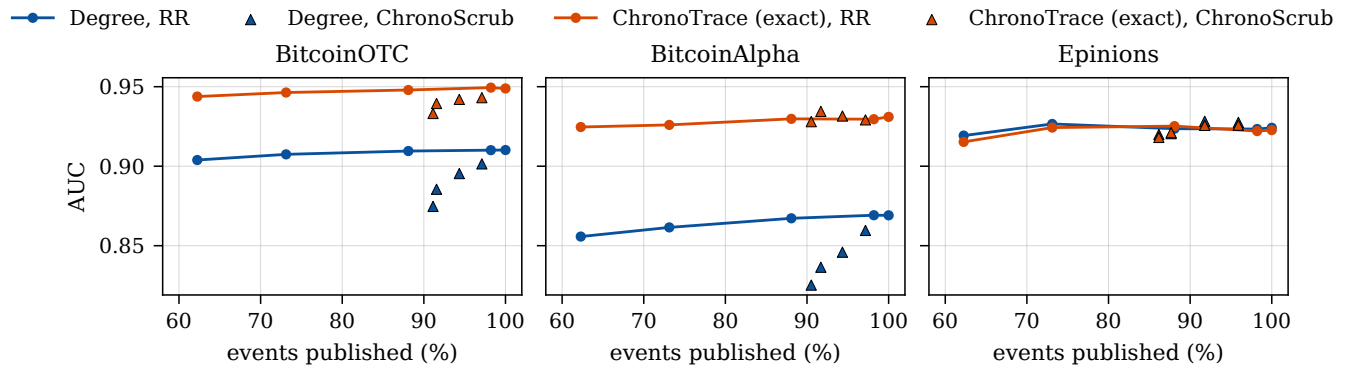


Figure 8: AUC against the share of events published, under randomized response (RR; $\epsilon = 4, 2, 1, 0.5$ from right to left) and the four ChronoScrub variants.

unchanged when candidates are subsampled, but AP, P@k and the best rank degrade as the pool grows: P@5 of the Degree adversary falls from 0.34 with 2,000 candidates to 0.10 with all 56,498, so results reported on smaller pools overstate operational precision (Appendix E). Third, a random-forest attacker ranks no better (Appendix D.9), and the sign-binarization threshold does not change the results (Appendix D.10). Fourth, node embeddings from a variant of DGTEN, a temporal trust GNN, rank former neighbors worse than degree even though they are learned before erasure (Appendix D.11).

Table 9: Adapted link-prediction heuristics (mean±std over 60 targets). R: release-only; W: reads the exact window and the number of former neighbors.

		BitcoinOTC			BitcoinAlpha			Epinions		
Heuristic		AUC	AP	R@500	AUC	AP	R@500	AUC	AP	R@500
Neighbor-count prior	R	0.907±0.086	0.104	0.742	0.865±0.126	0.084	0.714	0.922±0.083	0.052	0.391
|Sign balance|	R	0.283±0.095	0.001	0.024	0.280±0.133	0.001	0.011	0.208±0.103	0.000	0.007
Preferential attachment	W	0.907±0.086	0.104	0.742	0.865±0.126	0.084	0.714	0.922±0.083	0.052	0.391
Common neighbors	W	0.873±0.115	0.079	0.704	0.812±0.179	0.057	0.641	0.860±0.118	0.028	0.383
Jaccard	W	0.755±0.117	0.020	0.288	0.740±0.171	0.032	0.386	0.815±0.109	0.003	0.041
Adamic–Adar	W	0.879±0.115	0.099	0.731	0.821±0.178	0.071	0.660	0.862±0.118	0.033	0.393
Window activity share	W	0.773±0.130	0.004	0.156	0.747±0.152	0.006	0.283	0.580±0.187	0.000	0.002
Activity shift at window	W	0.758±0.146	0.041	0.513	0.699±0.176	0.042	0.465	0.775±0.153	0.013	0.240
Signed motif	W	0.922±0.070	0.073	0.757	0.898±0.090	0.060	0.728	0.880±0.115	0.032	0.358
CT-Unsupervised	W	0.929±0.061	0.057	0.774	0.905±0.085	0.058	0.776	0.917±0.087	0.017	0.295

7 Discussion

What platforms and custodians can do. Erasure leaves the partners' popularity in place, and popularity already points to likely partners: on all three graphs, degree alone accounts for the accuracy of every window-free adversary (Section 6.2). Deleting more edges around the erased user does not help (Section 6.5), but a platform can measure the risk by running ChronoTrace's calibration on its own graph. If the degree reference reproduces most of the adversary's advantage over chance, as on all three of our graphs, the residual risk is the popularity of the user's partners, which we expect only degree-level perturbation of the published graph to reduce. Two cheaper measures follow directly from our findings. A platform that keeps accounts visible should hide users left without events after an erasure, because they disclose former neighbors exactly. And a platform or custodian should not number the users of a release in order of first activity, because sequential IDs supply timing knowledge that adds signal beyond degree.

When the findings transfer. Across our three graphs, the AUC of the Degree adversary orders with degree heterogeneity (Table 3): the more unequal the degrees, the more a degree ranking reveals. The window-informed results depend on whether an edge records joint activity, as the quickly reciprocated Bitcoin ratings do (Section 6.3) and unilateral endorsements, such as Epinions trust statements, do not. We expect other marketplaces with mutual feedback, such as peer-to-peer lending or ride sharing, to resemble the Bitcoin graphs, and follower or endorsement graphs to resemble Epinions.

Operational meaning. An adversary with the exact window narrows the candidates sharply but still needs further evidence: the share of former neighbors among its top five candidates is at most 0.19, and on BitcoinOTC the best-ranked former neighbor sits at median rank 4 (mean 20). For an investigator, following up a top-five list in which at least 81% of the candidates are not former neighbors would be expensive. Combining the ranking with other sources, such as a second platform or known events, is a realistic escalation that we leave to future work.

8 Limitations

The adversary attacks one erasure in one published graph; several erasures, or two snapshots, would reveal more. The exact window is an upper bound, and only two estimators of it are evaluated. The adversaries are a logistic regression and a random forest on hand-built features, and the GNN embeddings we test were trained for sign prediction; an attacker that learns its own features for this task might find more. The three datasets come from two domains, all with binary signs; on the Bitcoin graphs, whose ratings range from -10 to $+10$, the binarization threshold does not change the results (Appendix D.10). Dated Epinions keeps only the statements with a genuine date, and our targets exclude users with more than 128 events. The reference graphs preserve degrees or activity but not reciprocity or community structure. On the static Epinions snapshot, structure ranks candidates of equal degree slightly above chance (Appendix E), so the finding that degree explains window-free inference may not hold for every graph. Each dataset contributes 60 targets drawn over degree quartiles, which bounds the precision of small differences; the confidence intervals report this. ChronoScrub is a diagnostic, the randomized response we evaluate only deletes events, and we did not evaluate defenses that perturb degrees. The secondary analyses of sign and time each use a single predictor.

9 Conclusion

Erasing a user from a trust graph removes the user's ratings, while the counterparties' ratings still reveal part of the user's relationships. ChronoTrace measures this residual inference and attributes it. An adversary without timing knowledge ranks former neighbors well because they are well connected, and calibration shows that degree explains all of it. Knowing when the user was active reaches relationships beyond degree, including those with low-degree users, as long as the window covers the user's active months, and mainly on graphs that record joint activity; sequential pseudonyms can supply that timing from the published graph itself. Platforms should hide accounts left isolated by an erasure, avoid sequential pseudonyms, and use calibration to decide whether degree-level perturbation is needed.

Ethical Considerations

This work uses only public, previously released network datasets and involves no human subjects, so it was not submitted to an institutional review board. The BitcoinOTC and BitcoinAlpha graphs are distributed through the SNAP repository [17], and the dated Epinions statements through the Network Repository [21]. All three contain only pseudonymous numeric identifiers and no names, contact details or content. We contacted no one, collected no new data, and report only aggregate metrics; no per-user result identifies an account. The method is dual-use. An investigator or a stalker could use the ranking to find associates of an erased user, and a platform or custodian can use the same ranking to audit its erasure procedure. Three factors bound the incremental harm. Precision is low (P@5 at most 0.19). Without timing knowledge, the adversary learns nothing beyond degree, which any public degree list already reveals. The two side channels we identify, isolated accounts and sequential pseudonyms, are cheap for platforms to close, and stating them helps platforms close them. Following the Menlo Report principles of beneficence and minimal risk [8], we judge publication to be net beneficial.

Open Science

The complete ChronoTrace benchmark is available at https://anonymous.4open.science/r/ChronoTrace-0265/. It contains feature extraction, the adversaries and their training on simulated erasures, the window estimators, the reference graphs and degree controls, the defenses, every per-target result file, and the scripts that produce every table, figure and number in this paper from those files. The datasets are public and are not redistributed: the artifact documents how to obtain them, and a preparation script rebuilds the exact input files, including the dated Epinions statements, and verifies them against checksums. The artifact will be submitted to the PoPETs artifact evaluation.

AI Use

The authors used LLM-based assistants to draft and edit text of this paper, to organize its LaTeX source, and to implement and test analysis code for this revision. The authors designed the study, checked the code and the results, and are responsible for every statement in the paper.

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A Feature Definitions

For candidate v in G_u^{rel}, let p_v and n_v be its numbers of positive and negative events, o_v and i_v its numbers of events given and received, and $h_v \in \mathbb{N}^{32}$ its events per time bin. (0) $\log(1+\deg v)$, where $\deg v$ counts v's events. (1) $(p_v - n_v)/\max(1, p_v + n_v)$. (2) $(o_v - i_v)/\max(1, o_v + i_v)$. (3) The share of v's events in $[b_0, b_1]$. (4) $|\bar{h}_v^{\mathrm{post}} - \bar{h}_v^{\mathrm{pre}}|$, the change in mean events per bin between the three bins after b_1 and the three bins before b_0, without normalization. (5, 6) The shares of v's positive and of its negative events in $[b_0, b_1]$. (7) $\log(1 +$ mean degree of v's neighbors$)$, zero without neighbors. (8) $\log(1 + |T_v|)$, where T_v is the set of users exactly two hops from v. (9) $|N(v)|/\sqrt{1+|T_v|}$, a closure ratio that is large when v's neighborhood closes on itself (it is not the clustering coefficient). (10) $1 - H(h_v)/\log 32$, where H is the Shannon entropy of v's normalized bin counts. All features are standardized before fitting.

B Experimental Protocol

This appendix specifies the protocol behind every result. The released code implements exactly these steps.

B.1 Data

Each dataset is a list of rating events (rater, rated user, rating, time), sorted by time. A rating above zero is positive and any other rating is negative; Bitcoin ratings are never zero, and every Epinions statement is either trust or distrust. We use the events as released, without merging repeated pairs (none occur) or renumbering users, with one exception. The dated Epinions graph keeps every statement of the Extended Epinions release [19, 21] that carries a genuine creation date, without self-ratings, and numbers its users in order of first appearance: users who rated others come first, ordered by their first rating given, followed by users who were only rated. BitcoinOTC and BitcoinAlpha keep the user IDs of their SNAP releases [17]. The static snapshot keeps its self-ratings and uses the record index as time. The 32 time bins divide the span from the first to the last event of each real graph into equal intervals, and the reference graphs use the same bins. Reciprocated ratings come back quickly on the Bitcoin graphs, where 86% and 84% of reciprocated pairs are reciprocated within a week, against 62% on Epinions.

B.2 Targets, candidates and labels

We use the seeds 42, 7, 13, 100 and 200. For each seed, the main targets are three users drawn at random from each quartile of the incident-event count among the users with 4 to 128 events. The low-degree targets of Appendix D.5 are 12 users per seed drawn uniformly among the users with one to three events. A user drawn under two seeds counts as two targets; the main targets contain 1 such repeat. In an edge-list release, the candidates are the users with at least one event in the published graph, after any defense; a former neighbor left without events is neither a positive nor a negative and is counted as isolated. In a directory release, every user except u is a candidate. A target with no former neighbor among its candidates has no defined AUC and is left out; this happens only among low-degree targets (Appendix D.5).

B.3 Training the adversaries

For a target u drawn with a given seed, the shadow users are the other main targets of that seed: eleven for a main target and twelve for a low-degree target. Each shadow erasure deletes a shadow user u_j from G_u^{rel}. Its labels are u_j's neighbors in G_u^{rel}, and its exact window comes from u_j's events in G_u^{rel}, so no event, label or window of u enters training. The adversary pools the rows of all shadow erasures and keeps every positive. From the n_- negatives it draws $\min(n_-, \max(1000, 20n_+))$ without replacement, where n_+ counts the positives, using a generator seeded by the seed plus u's ID. It standardizes the features on the drawn rows and fits a logistic regression with an L2 penalty ($C = 1$), balanced class weights and at most 1,000 iterations. The random forest of Appendix D.9 uses the same rows. The submitted protocol, compared in Appendix C.2, differs only in its training graphs: it deleted each shadow user from the original graph G, which still contains u.

B.4 Timing knowledge

The exact window spans the bins of u's first and last events in G, given or received, padded by one bin on each side and clipped to the 32 bins. A shift by δ moves the window by δ bins in a direction drawn at random for each target and keeps its width; near the ends of the time span the window

moves only as far as the span allows. Each shadow window is shifted by the same δ in its own random direction. The hub estimator ranks the users of the published graph by their number of distinct neighbors, takes the median time of all published events that touch one of the top 50, and returns the seven bins centered on that time, truncated at the ends of the span. The ID-cohort estimator takes the 25 users of the published graph whose IDs are closest to u's and finds each one's first and last active bins. It returns the bins from the floor of the median first bin to the ceiling of the median last bin, padded by one bin. Both estimators treat the shadow erasures in the same way.

B.5 Metrics and statistics

AUC counts ties as one half, and AP is the non-interpolated average precision. Ranks start at 1 for the top candidate. The degree-matched AUC is the AUC over all pairs of a former neighbor and a non-neighbor that have the same number of events in G_u^{rel}. All intervals are percentile bootstrap intervals over targets with 10,000 resamples, and paired differences use the targets present in both arms. The p-values are two-sided and come from 10,000 random sign flips of the per-target differences. The Benjamini–Hochberg adjustment applies within each dataset to the eight comparisons of Table 13; all other p-values are unadjusted. The groups by $|N_u|$ in Figure 7 are the tertiles of the number of former neighbors present in the release. The pool-size analysis of Appendix E reuses the scores computed on the full universe, keeps every former neighbor, and averages ten random draws of non-neighbors for each pool size.

B.6 Defenses and utility

The residue of ChronoScrub consists of N_u and every neighbor of N_u in G_u^{rel}. Each event of G_u^{rel} receives a drop score. At strength s, an event inside W_u that touches the residue scores $0.45s$, with $s = 1$ for the temporal variant. In the joint variant, an event between two residue users scores at least $0.25s$, at any time. A deterministic hash maps each event to a number in $[0, 1)$, and the event is deleted when that number is below its drop score. Randomized response keeps every event of G_u^{rel} independently with probability $e^{\epsilon}/(1+e^{\epsilon})$, using a generator seeded by the seed plus u's ID.

Utility is measured by sign prediction. The test set consists of the latest 20% of the events of the undefended release, of which at most 4,000 are sampled. The training set consists of the events of the defended release that precede them, of which at most 12,000 are sampled. Each event is described by eight features of its endpoints in the training graph, computed without the event itself: the logarithms of both endpoints' degrees, of the rater's out-degree and of the rated user's in-degree; both endpoints' sign balances; the logarithm of their number of common neighbors; and their Jaccard similarity. A logistic regression with balanced class weights predicts the sign, and utility is its AUC on the test set.

Table 10: Reference graphs: rewirings per dataset, and mean numbers of events dropped as self-loops and of repeated rater–rated pairs per rewiring.

Dataset	Reference	Rewirings	Self-loops	Duplicates
BitcoinOTC	degree	5	0	1539
BitcoinOTC	activity	5	0	2528
BitcoinAlpha	degree	5	0	1076
BitcoinAlpha	activity	5	0	1753
Epinions	degree	3	0	9965
Epinions	activity	3	0	13565

B.7 Sign and time of deleted relationships

Both analyses use every pair of a main target u and a former neighbor v that remains a candidate in the edge-list release. The relationship counts as distrust if any rating between u and v, in either direction, is negative. The sign predictor scores v by the negative of its sign balance in G_u^{rel} (feature 1), and the AUC pools all pairs. The time of the relationship is the bin of the earliest event between u and v. We compare three predictors of it: the dataset's busiest bin over all events of G; v's busiest bin in G_u^{rel}; and, for a window-informed adversary, v's busiest bin inside W_u. Ties go to the earliest bin, and intervals bootstrap over targets.

B.8 Computation

All edge-list experiments together take about four hours of wall-clock time on up to 22 CPU worker processes. The GNN of Appendix D.11 trains in minutes on one GPU.

C Calibration Details

C.1 Reference graphs

Both rewirings keep each event's rater, sign and time, and permute the rated users within groups: over all events for the degree reference, and among the events of the same time bin and sign for the activity reference. A permutation can map an event onto its own rater. We repair such self-loops by swapping rated users within the group and would drop any that remained; none did. Repeated rater–rated pairs are kept, so the rewiring preserves every user's event counts but neither its number of distinct neighbors nor the graph's reciprocity. We use five rewirings of each Bitcoin graph and three of Epinions, seeded identically for both references, and average each target's result over them. Table 10 reports the number of rewirings and the dropped and duplicated events. The label reference draws $|N_u|$ candidates without replacement with probability proportional to their degree in G_u^{rel}, 200 times per target, and scores the Degree ranking against them.

C.2 Protocol check

Table 11 compares the two ways of training the adversary. The submitted protocol trained on erasures of other targets from the original graph, so its training graphs still contained the evaluated target and its edges. The protocol used throughout this revision trains only on erasures simulated on

the published graph. Every difference is below 0.001 AUC in both release formats, so the target's own information did not drive the results.

D Extended Results

This appendix gives every result behind the main text, organized by the question it supports.

D.1 Ranking metrics at fixed depths

Table 12 complements Table 4 with precision and success at the top of the ranking and with rank statistics, and Figure 9 plots P@k and R@k. ChronoTrace with the exact window leads or ties in precision and rank on the Bitcoin graphs, although Sgn-Struct has a slightly higher S@10. On Epinions, Degree has the highest P@5 although the learned sets have similar AUC, because their gains come from the middle of the ranking rather than its top.

D.2 Paired comparisons

Table 13 lists every paired comparison against the Degree adversary with bootstrap intervals and Benjamini–Hochberg adjusted p-values. It is the numerical counterpart of Figure 4.

D.3 Per-target heterogeneity

Figure 10 shows every target's AUC against its number of former neighbors. Exposure varies widely between users with few former neighbors and less between users with many.

D.4 Timing knowledge

Table 14 reports ChronoTrace under every level of timing knowledge, with the overlap of each window with the exact one. Table 15 repeats the analysis for all three window-based feature sets. The pattern of Figure 6 holds for each of them: a one-bin shift changes nothing, a two-bin shift brings every set down to about the Degree adversary, and an eight-bin shift leaves Temporal, which has no degree feature, well below it (0.847 and 0.773 on the Bitcoin graphs).

D.5 Low-degree targets

Many users have only one to three events, so we also erase 60 such users per dataset, 12 per seed, drawn uniformly. The adversary trains on simulated erasures of the same seed's main targets, which supply more positive examples than low-degree users would. A target whose former neighbors all disappear from the edge-list release has no defined AUC and is left out of the averages; in a directory release those neighbors would be identified exactly. Table 16 reports the results.

D.6 Sign and time of deleted relationships

Table 17 gives the numbers behind Section 6.4: for every dataset, the share of pairs whose relationship included distrust, the AUC of the sign predictor with its confidence interval, and the accuracy of every time predictor of Appendix B.7.

D.7 Rare former neighbors

Table 7 groups former neighbors by degree. Table 18 summarizes the same question in one number per adversary, the surprisal-weighted recall, in which rare former neighbors count more. It weights each former neighbor v by $-\log q_v$, where $q_v = (\deg v + 1)/\sum_{w \in C_u} (\deg w + 1)$. The weighting lowers every adversary's recall, because rare neighbors are harder to find. On the Bitcoin graphs it costs ChronoTrace with the exact window less than Degree: at depth 100, Degree loses 14% and 14% of its recall, and ChronoTrace 10% and 7%.

D.8 Adapted heuristics

Figure 11 plots the heuristics of Table 9 with their confidence intervals. Preferential attachment and the neighbor-count prior produce identical rankings: preferential attachment is $|N(v)| \cdot \bar{n}_A$, where the anchors' mean neighbor count $\bar{n}_A$ is constant within a target, and the prior $\log(1 + |N(v)|)$ is monotone in $|N(v)|$. Both rank by neighbor count, whereas the Degree adversary ranks by degree, the number of events; the two differ when ratings are reciprocated, which explains their slightly different results.

D.9 Classifier family

A random forest (100 trees, depth 10, balanced class weights) trained on the same simulated erasures does not rank better than the logistic adversary for any feature set (Table 19). On Degree alone it ranks much worse, so it is a weaker attacker than the logistic regression rather than a stronger one.

D.10 Sign binarization

Bitcoin ratings range from -10 to $+10$ and are never zero, so the thresholds $\tau = -1$ and $\tau = 0$ give identical signs. Table 20 varies the threshold above which a rating counts as positive; ChronoTrace's AUC barely changes on either Bitcoin graph. Epinions statements carry only a sign.

D.11 Temporal trust GNN embeddings

DGTEN [25] is a temporal trust-prediction graph neural network. We use a deterministic variant without its uncertainty modeling: three signed graph-convolution layers followed by a temporal attention module. It is trained for 50 epochs on the first seven of ten equal-length time snapshots of each Bitcoin graph before erasure, with input features smoothed over the whole graph, and we append its 32-dimensional node embeddings to the candidate features. The embeddings are learned from the graph before erasure, which contains the deleted edges, so this baseline sees more than a release-only adversary may. Even so, the embeddings rank former neighbors worse than degree, and adding them to the hand-engineered features does not help (Table 21). The embeddings are optimized for sign prediction, not for neighbor inference, so this result does not show that graph neural networks are unsuitable for the task.

Table 11: AUC with training on erasures simulated on the published graph minus AUC with the submitted protocol, in both release formats (mean [95% CI] over 60 targets).

Release	Feature set	BitcoinOTC	BitcoinAlpha	Epinions
edge-list	Degree	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]
	Topology	−0.0001 [−0.0006, +0.0002]	+0.0002 [−0.0003, +0.0007]	+0.0003 [+0.0001, +0.0006]
	Full-R	−0.0003 [−0.0010, +0.0003]	−0.0001 [−0.0007, +0.0005]	−0.0001 [−0.0006, +0.0004]
	Temporal	−0.0006 [−0.0016, 0.0000]	−0.0001 [−0.0003, +0.0002]	+0.0002 [−0.0001, +0.0006]
	Sgn-Struct	−0.0004 [−0.0017, +0.0004]	−0.0001 [−0.0009, +0.0005]	+0.0004 [−0.0001, +0.0011]
	ChronoTrace (exact)	−0.0001 [−0.0010, +0.0008]	−0.0001 [−0.0010, +0.0009]	+0.0004 [−0.0001, +0.0010]
directory	Degree	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]
	Topology	−0.0003 [−0.0008, +0.0001]	+0.0002 [−0.0003, +0.0007]	−0.0004 [−0.0008, 0.0000]
	Full-R	−0.0004 [−0.0009, 0.0000]	−0.0001 [−0.0007, +0.0006]	+0.0001 [−0.0003, +0.0005]
	Temporal	−0.0001 [−0.0003, +0.0001]	−0.0008 [−0.0023, +0.0001]	+0.0001 [−0.0001, +0.0003]
	Sgn-Struct	−0.0004 [−0.0008, 0.0000]	+0.0001 [−0.0005, +0.0007]	−0.0001 [−0.0006, +0.0004]
	ChronoTrace (exact)	−0.0002 [−0.0009, +0.0005]	+0.0001 [−0.0008, +0.0010]	−0.0002 [−0.0008, +0.0004]

Table 12: Ranking metrics at fixed depths (mean over 60 targets). S@k: at least one former neighbor in the top k. Best and mean: rank of the best-ranked and the average rank of the former neighbors.

Dataset	Feature set	Window	P@5	P@10	P@20	S@5	S@10	Best	Mean
BitcoinOTC	Degree	none	0.13	0.11	0.08	0.48	0.68	35	533
	Topology	none	0.11	0.10	0.08	0.38	0.67	35	547
	Full-R	none	0.09	0.09	0.07	0.37	0.58	36	589
	ChronoTrace	hub (from release)	0.08	0.09	0.07	0.27	0.52	38	597
	ChronoTrace	ID cohort (from release)	0.11	0.10	0.08	0.42	0.62	23	426
	Temporal	exact	0.13	0.11	0.08	0.53	0.63	29	361
	Sgn-Struct	exact	0.16	0.13	0.10	0.58	0.73	33	323
	ChronoTrace	exact	0.19	0.14	0.10	0.58	0.72	20	305
BitcoinAlpha	Degree	none	0.08	0.07	0.06	0.37	0.50	67	494
	Topology	none	0.07	0.07	0.05	0.28	0.48	67	491
	Full-R	none	0.07	0.06	0.05	0.30	0.40	65	503
	ChronoTrace	hub (from release)	0.05	0.05	0.05	0.22	0.35	75	546
	ChronoTrace	ID cohort (from release)	0.07	0.06	0.05	0.25	0.45	73	456
	Temporal	exact	0.09	0.08	0.07	0.42	0.53	53	366
	Sgn-Struct	exact	0.11	0.09	0.08	0.43	0.60	30	267
	ChronoTrace	exact	0.11	0.09	0.08	0.43	0.58	30	264
Epinions	Degree	none	0.10	0.08	0.06	0.33	0.35	167	4246
	Topology	none	0.05	0.04	0.04	0.18	0.23	172	4195
	Full-R	none	0.05	0.04	0.04	0.18	0.27	182	4416
	ChronoTrace	hub (from release)	0.04	0.04	0.03	0.15	0.27	191	4572
	ChronoTrace	ID cohort (from release)	0.04	0.04	0.03	0.13	0.22	180	4194
	Temporal	exact	0.04	0.04	0.03	0.17	0.27	635	5608
	Sgn-Struct	exact	0.08	0.07	0.06	0.27	0.35	129	4189
	ChronoTrace	exact	0.04	0.04	0.04	0.18	0.22	160	4374

Table 13: Paired differences in AUC against the Degree adversary (last row of each block: against Full-R), with 95% bootstrap CIs over 60 targets, two-sided sign-flip p-values, and Benjamini–Hochberg adjusted p-values within each dataset.

Dataset	Comparison	ΔAUC	95% CI	p	p_{BH}
BitcoinOTC	Topology	−0.002	[−0.006, +0.001]	0.235	0.235
	Full-R	−0.010	[−0.019, −0.002]	0.025	0.029
	ChronoTrace, hub window	−0.011	[−0.021, −0.002]	0.021	0.029
	ChronoTrace, ID-cohort window	+0.018	[+0.003, +0.031]	0.010	0.016
	Temporal	+0.029	[+0.014, +0.044]	<0.001	<0.001
	Sgn-Struct	+0.036	[+0.022, +0.050]	<0.001	<0.001
	ChronoTrace, exact window	+0.039	[+0.026, +0.053]	<0.001	<0.001
	ChronoTrace (exact) minus Full-R	+0.048	[+0.034, +0.064]	<0.001	<0.001
BitcoinAlpha	Topology	+0.002	[−0.006, +0.010]	0.630	0.720
	Full-R	−0.001	[−0.010, +0.008]	0.776	0.776
	ChronoTrace, hub window	−0.013	[−0.033, +0.005]	0.212	0.339
	ChronoTrace, ID-cohort window	+0.011	[−0.010, +0.033]	0.324	0.432
	Temporal	+0.038	[+0.014, +0.062]	0.002	0.004
	Sgn-Struct	+0.061	[+0.043, +0.081]	<0.001	<0.001
	ChronoTrace, exact window	+0.062	[+0.043, +0.083]	<0.001	<0.001
	ChronoTrace (exact) minus Full-R	+0.063	[+0.046, +0.081]	<0.001	<0.001
Epinions	Topology	+0.002	[−0.003, +0.006]	0.457	0.915
	Full-R	−0.002	[−0.007, +0.003]	0.438	0.915
	ChronoTrace, hub window	−0.005	[−0.013, +0.002]	0.223	0.892
	ChronoTrace, ID-cohort window	+0.002	[−0.006, +0.009]	0.656	0.958
	Temporal	−0.021	[−0.037, −0.006]	0.008	0.066
	Sgn-Struct	+0.002	[−0.008, +0.012]	0.728	0.958
	ChronoTrace, exact window	−0.001	[−0.017, +0.012]	0.861	0.958
	ChronoTrace (exact) minus Full-R	0.000	[−0.014, +0.013]	0.958	0.958

Table 14: ChronoTrace under each level of timing knowledge (mean over 60 targets). IoU: overlap with the exact window. Δ: paired difference to the Degree adversary. Excess: excess over degree.

Dataset	Window	IoU	AUC [95% CI]	Δ vs Degree [95% CI]	DM-AUC	Excess
BitcoinOTC	exact	1.00	0.949 [0.937, 0.960]	+0.039 [+0.026, +0.053]	0.81	+0.079*
	shifted, $\delta = 1$	0.66	0.948 [0.937, 0.958]	+0.038 [+0.024, +0.052]	0.80	+0.077*
	shifted, $\delta = 2$	0.43	0.911 [0.889, 0.932]	+0.001 [−0.014, +0.015]	0.63	+0.026*
	shifted, $\delta = 4$	0.22	0.906 [0.881, 0.929]	−0.004 [−0.021, +0.011]	0.62	+0.020*
	shifted, $\delta = 8$	0.12	0.893 [0.870, 0.915]	−0.017 [−0.029, −0.006]	0.49	−0.003
	ID cohort	0.66	0.928 [0.911, 0.944]	+0.018 [+0.003, +0.031]	0.70	+0.040*
	hub	0.26	0.899 [0.875, 0.921]	−0.011 [−0.021, −0.002]	0.55	+0.002
BitcoinAlpha	exact	1.00	0.931 [0.914, 0.947]	+0.062 [+0.043, +0.083]	0.81	+0.109*
	shifted, $\delta = 1$	0.68	0.931 [0.913, 0.946]	+0.061 [+0.042, +0.082]	0.81	+0.107*
	shifted, $\delta = 2$	0.46	0.890 [0.862, 0.916]	+0.021 [+0.009, +0.035]	0.72	+0.051*
	shifted, $\delta = 4$	0.30	0.881 [0.853, 0.907]	+0.011 [−0.006, +0.027]	0.68	+0.045*
	shifted, $\delta = 8$	0.16	0.861 [0.832, 0.889]	−0.008 [−0.017, +0.002]	0.50	+0.004
	ID cohort	0.53	0.880 [0.847, 0.909]	+0.011 [−0.010, +0.033]	0.57	+0.040*
	hub	0.22	0.856 [0.822, 0.888]	−0.013 [−0.033, +0.005]	0.61	+0.007
Epinions	exact	1.00	0.923 [0.899, 0.944]	−0.001 [−0.017, +0.012]	0.59	+0.011
	shifted, $\delta = 1$	0.80	0.924 [0.901, 0.945]	0.000 [−0.016, +0.013]	0.61	+0.012
	shifted, $\delta = 2$	0.67	0.921 [0.898, 0.941]	−0.003 [−0.019, +0.009]	0.55	+0.007
	shifted, $\delta = 4$	0.53	0.915 [0.890, 0.938]	−0.009 [−0.024, +0.001]	0.49	0.000
	shifted, $\delta = 8$	0.47	0.920 [0.898, 0.940]	−0.004 [−0.011, +0.004]	0.44	+0.005
	ID cohort	0.48	0.926 [0.905, 0.945]	+0.002 [−0.006, +0.009]	0.59	+0.011*
	hub	0.22	0.919 [0.898, 0.939]	−0.005 [−0.013, +0.002]	0.48	+0.001

* $p < 0.05$, two-sided sign-flip permutation test (unadjusted).

Table 15: AUC of the three window-based feature sets under each level of timing knowledge (mean over 60 targets); first row: the Degree adversary.

	BitcoinOTC			BitcoinAlpha			Epinions		
Window	Temporal	Sgn-Struct	ChronoTrace	Temporal	Sgn-Struct	ChronoTrace	Temporal	Sgn-Struct	ChronoTrace
none (Degree)		0.910			0.869			0.924	
exact	0.939	0.946	0.949	0.907	0.930	0.931	0.903	0.926	0.923
shifted, $\delta = 1$	0.939	0.944	0.948	0.908	0.930	0.931	0.907	0.928	0.924
shifted, $\delta = 2$	0.914	0.905	0.911	0.879	0.876	0.890	0.904	0.924	0.921
shifted, $\delta = 4$	0.908	0.900	0.906	0.874	0.865	0.881	0.901	0.917	0.915
shifted, $\delta = 8$	0.847	0.897	0.893	0.773	0.860	0.861	0.888	0.920	0.920
ID cohort	0.921	0.927	0.928	0.837	0.880	0.880	0.904	0.925	0.926
hub	0.862	0.902	0.899	0.787	0.861	0.856	0.886	0.921	0.919

Table 16: Targets with one to three events (60 drawn per dataset). Best: mean rank of the best-ranked former neighbor. n: targets with at least one former neighbor in the edge-list release.

Dataset	Feature set	AUC [95% CI]	S@10	Best	n
BitcoinOTC	Degree	0.953 [0.925, 0.975]	0.32	187	59
	Full-R	0.954 [0.928, 0.974]	0.29	198	59
	ChronoTrace, ID-cohort window	0.971 [0.954, 0.984]	0.31	121	59
	ChronoTrace, exact window	0.978 [0.965, 0.988]	0.36	99	59
BitcoinAlpha	Degree	0.893 [0.845, 0.932]	0.27	333	60
	Full-R	0.901 [0.857, 0.937]	0.20	317	60
	ChronoTrace, ID-cohort window	0.913 [0.875, 0.944]	0.25	273	60
	ChronoTrace, exact window	0.956 [0.938, 0.972]	0.23	138	60
Epinions	Degree	0.903 [0.863, 0.935]	0.04	4316	46
	Full-R	0.898 [0.849, 0.937]	0.09	4401	46
	ChronoTrace, ID-cohort window	0.901 [0.848, 0.940]	0.09	4311	46
	ChronoTrace, exact window	0.906 [0.859, 0.942]	0.09	3677	46

Table 17: Sign and time of deleted relationships, over pairs of a main target and a former neighbor present in the edge-list release. Distrust: share of pairs with a negative rating. Mode: the dataset's busiest bin; Busiest: the neighbor's busiest bin; W_u: exact window.

				Same bin				Within one bin			
Dataset	Pairs	Distrust	Sign AUC [95% CI]	Mode	Busiest	Busiest in W_u	Middle of W_u	Mode	Busiest	Busiest in W_u	Middle of W_u
BitcoinOTC	543	14%	0.82 [0.76, 0.88]	9%	43%	50%	29%	14%	61%	75%	56%
BitcoinAlpha	443	5%	0.82 [0.72, 0.92]	26%	47%	50%	29%	30%	69%	80%	50%
Epinions	968	23%	0.86 [0.83, 0.89]	4%	31%	36%	16%	19%	47%	62%	31%

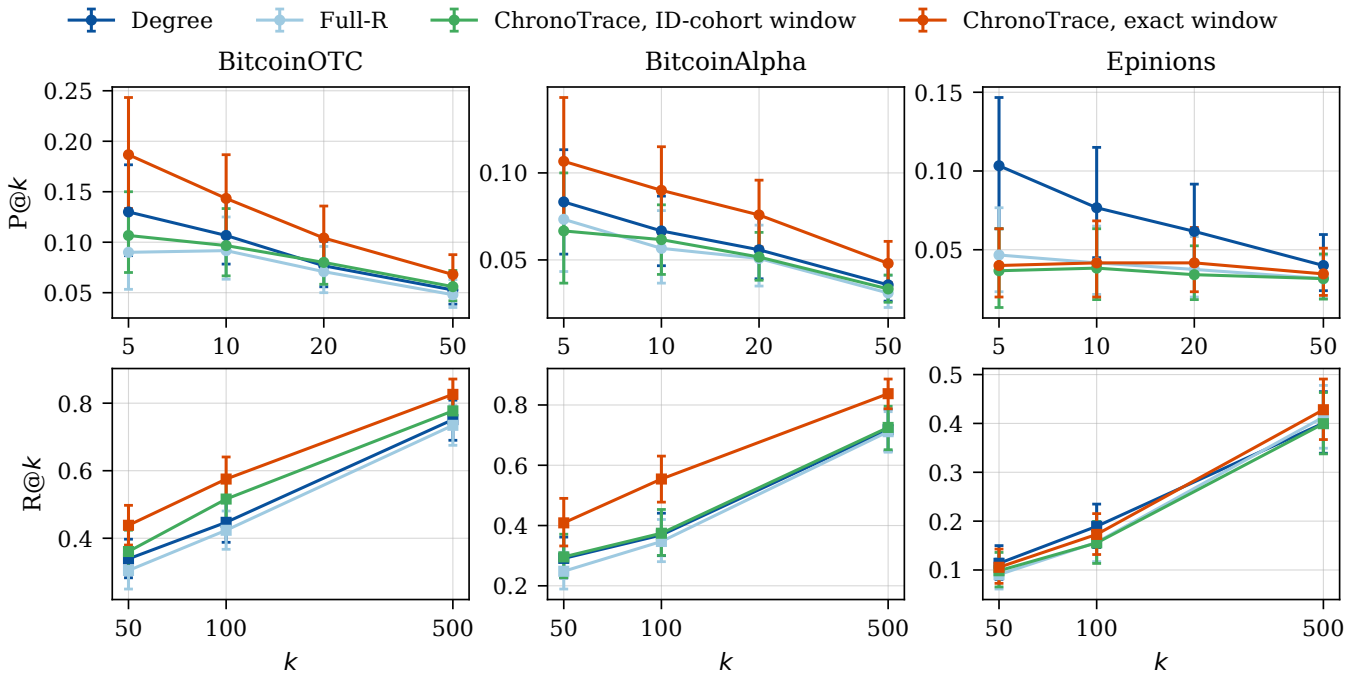


Figure 9: Precision (top) and recall (bottom) at depth k (95% CI over 60 targets).

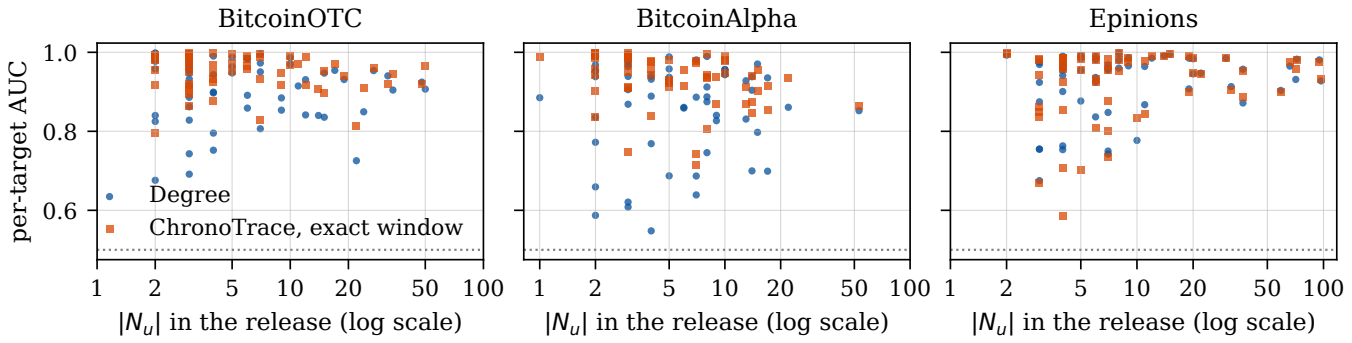


Figure 10: Per-target AUC of the Degree and ChronoTrace (exact window) adversaries against $|N_u|$.

Table 18: Recall (R) and surprisal-weighted recall (SWR) at depths 100 and 500 (mean over 60 targets).

Dataset	Feature set	Window	R@100	SWR@100	R@500	SWR@500
BitcoinOTC	Degree	none	0.447	0.384	0.751	0.705
	Full-R	none	0.424	0.360	0.734	0.686
	ChronoTrace	ID cohort	0.516	0.452	0.777	0.735
	ChronoTrace	exact	0.575	0.517	0.826	0.791
BitcoinAlpha	Degree	none	0.369	0.318	0.718	0.680
	Full-R	none	0.347	0.300	0.711	0.672
	ChronoTrace	ID cohort	0.374	0.333	0.726	0.694
	ChronoTrace	exact	0.554	0.514	0.837	0.812
Epinions	Degree	none	0.189	0.150	0.401	0.343
	Full-R	none	0.157	0.127	0.413	0.355
	ChronoTrace	ID cohort	0.155	0.124	0.399	0.345
	ChronoTrace	exact	0.173	0.140	0.428	0.370

D.12 Defenses in full

Table 22 adds standard deviations, recall at depth 500, and the spread of the share of events kept to Table 8. Sign prediction from endpoint features is weak on BitcoinAlpha, so its utility column says less than those of the other datasets.

E Continuity with the Submitted Evaluation

Besides its training protocol (Appendix C.2), the submitted version differed from this revision in three ways: it evaluated directory releases, it subsampled 5,000 candidates on BitcoinOTC and Epinions, and it used the static SNAP Epinions snapshot. This appendix reports each of the three settings.

Table 19: Logistic regression (LR) and random forest (RF) attackers trained on the same simulated erasures (mean AUC over 60 targets).

	Degree		Full-R		ChronoTrace (exact)	
Dataset	LR	RF	LR	RF	LR	RF
BitcoinOTC	0.910	0.756	0.901	0.866	0.949	0.929
BitcoinAlpha	0.869	0.725	0.868	0.838	0.931	0.903
Epinions	0.924	0.852	0.922	0.888	0.923	0.897

Table 20: ChronoTrace (exact window) when ratings above τ count as positive (mean AUC over 60 targets).

Dataset	$\tau = -1$	$\tau = 0$	$\tau = 1$	$\tau = 5$	Distrust at $\tau = 0$
BitcoinOTC	0.949	0.949	0.950	0.949	10%
BitcoinAlpha	0.931	0.931	0.928	0.929	6%

Table 21: Node embeddings of a DGTEN variant as attacker features (mean±std over 60 targets; Δ: paired difference to Degree with 95% CI).

Dataset	Features	AUC	Δ vs Degree
BitcoinOTC	Degree	0.910±0.081	–
	DGTEN embedding	0.833±0.133	−0.077 [−0.107, −0.051]
	Full-R + DGTEN	0.869±0.122	−0.042 [−0.070, −0.021]
	ChronoTrace (exact)	0.949±0.047	+0.039 [+0.026, +0.053]
	ChronoTrace (exact) + DGTEN	0.931±0.076	+0.020 [−0.001, +0.040]
BitcoinAlpha	Degree	0.869±0.119	–
	DGTEN embedding	0.752±0.199	−0.117 [−0.157, −0.078]
	Full-R + DGTEN	0.800±0.171	−0.069 [−0.103, −0.039]
	ChronoTrace (exact)	0.931±0.066	+0.062 [+0.043, +0.083]
	ChronoTrace (exact) + DGTEN	0.887±0.106	+0.018 [−0.010, +0.046]

Table 22: Defenses in full (mean±std over 60 targets): AUC of both adversaries, R@500 of ChronoTrace, share of events kept, and sign-prediction utility. RR: randomized response.

Dataset	Defense	Degree AUC	ChronoTrace AUC	ChronoTrace R@500	Kept	Utility
BitcoinOTC	None	0.910±0.081	0.949±0.047	0.826	1.00±0.00	0.635±0.012
	ChronoScrub, temporal	0.875±0.117	0.933±0.070	0.792	0.91±0.07	0.632±0.016
	ChronoScrub, joint 0.25	0.901±0.091	0.943±0.057	0.802	0.97±0.02	0.636±0.014
	ChronoScrub, joint 0.50	0.895±0.097	0.942±0.057	0.793	0.94±0.04	0.632±0.013
	ChronoScrub, joint 0.75	0.886±0.111	0.939±0.061	0.823	0.92±0.06	0.633±0.011
	RR, $\epsilon = 0.5$	0.904±0.087	0.944±0.052	0.813	0.62±0.00	0.639±0.014
	RR, $\epsilon = 1$	0.907±0.086	0.946±0.050	0.826	0.73±0.00	0.637±0.012
	RR, $\epsilon = 2$	0.910±0.080	0.948±0.048	0.810	0.88±0.00	0.637±0.013
	RR, $\epsilon = 4$	0.910±0.081	0.949±0.046	0.813	0.98±0.00	0.639±0.018
BitcoinAlpha	None	0.869±0.119	0.931±0.066	0.837	1.00±0.00	0.545±0.012
	ChronoScrub, temporal	0.825±0.144	0.928±0.070	0.829	0.91±0.09	0.544±0.011
	ChronoScrub, joint 0.25	0.860±0.125	0.929±0.069	0.842	0.97±0.02	0.545±0.011
	ChronoScrub, joint 0.50	0.846±0.137	0.932±0.069	0.837	0.94±0.05	0.549±0.011
	ChronoScrub, joint 0.75	0.836±0.137	0.934±0.070	0.866	0.92±0.07	0.550±0.010
	RR, $\epsilon = 0.5$	0.856±0.130	0.925±0.075	0.854	0.62±0.00	0.544±0.011
	RR, $\epsilon = 1$	0.861±0.127	0.926±0.078	0.856	0.73±0.00	0.544±0.011
	RR, $\epsilon = 2$	0.867±0.120	0.930±0.074	0.840	0.88±0.00	0.547±0.011
	RR, $\epsilon = 4$	0.869±0.119	0.930±0.067	0.836	0.98±0.00	0.548±0.010
Epinions	None	0.924±0.081	0.923±0.092	0.428	1.00±0.00	0.767±0.013
	ChronoScrub, temporal	0.920±0.081	0.918±0.091	0.387	0.86±0.11	0.768±0.014
	ChronoScrub, joint 0.25	0.928±0.077	0.926±0.091	0.425	0.96±0.03	0.766±0.012
	ChronoScrub, joint 0.50	0.928±0.077	0.926±0.091	0.404	0.92±0.05	0.766±0.013
	ChronoScrub, joint 0.75	0.920±0.083	0.921±0.087	0.381	0.88±0.08	0.767±0.013
	RR, $\epsilon = 0.5$	0.919±0.081	0.915±0.095	0.434	0.62±0.00	0.767±0.014
	RR, $\epsilon = 1$	0.927±0.075	0.924±0.087	0.437	0.73±0.00	0.768±0.014
	RR, $\epsilon = 2$	0.924±0.082	0.925±0.088	0.459	0.88±0.00	0.764±0.015
	RR, $\epsilon = 4$	0.923±0.081	0.922±0.094	0.436	0.98±0.00	0.769±0.012

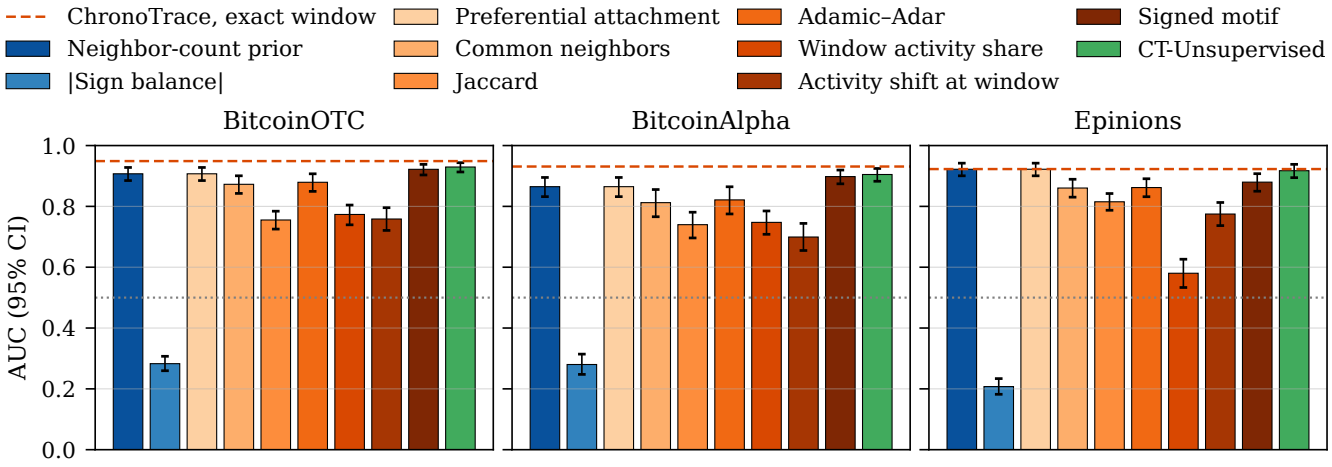


Figure 11: Adapted heuristics (AUC, 95% CI). Dashed: ChronoTrace with the exact window.

Directory releases. Table 23 reports the main adversaries in the directory setting. The isolated former neighbors lower the Degree adversary there. They produce the negative gaps to the degree reference that the submitted version reported and called deletion leakage, and the gains of structural features that it reported on BitcoinAlpha and Epinions.

Candidate pools. Table 24 and Figure 12 show how the ranking metrics depend on the size of the candidate pool. AUC does not change with the pool size, whereas AP, P@5, S@10 and the best rank degrade as more non-neighbors compete for the top positions.

Static Epinions snapshot. Table 25 evaluates the static snapshot on its full universe of 131,827 candidates. Its time column is the record order, and 97% of consecutive records have a rater ID at least as large as the previous record's, so the order largely follows rater IDs. The timing features therefore read file order, and ChronoTrace's gain of +0.014 (95% CI [+0.003, +0.026]) over Degree does not measure timing knowledge. Unlike on the three dated graphs, structure ranks candidates of equal degree slightly above chance on this snapshot: Topology has degree-matched AUC 0.54 and excess over degree +0.007, although its AUC gain over Degree, +0.003 (95% CI [−0.001, +0.006]), is not significant. Full-R ranks candidates of equal degree better still (degree-matched AUC 0.60), because its activity concentration (feature 10) reads record order.

F Reproducibility

Every number in this paper is a macro, and every data table and plot is a file, generated by two scripts from the per-target result files; no number is typed by hand. The released artifact (see Open Science) contains the code, every per-target record, the run logs and these scripts. A preparation script rebuilds the input files from the public releases and checks them against SHA-256 checksums of the files we used, and it reproduces all of them byte for byte. Every random choice is seeded: target sampling, negative sampling, window shifts, rewiring, label references, randomized response, the bootstrap and the permutation tests. With one BLAS thread per worker, rerunning the experiments reproduces the saved records; we checked this by rerunning one seed of BitcoinOTC from scratch, which gave the same values for every metric, coefficient and window of its 12 targets.

All experiments ran with Python 3.10, NumPy 1.26, SciPy 1.15, scikit-learn 1.6, pandas 2.3 and Matplotlib 3.10; the GNN baseline additionally used PyTorch 2.4. Regenerating every table and figure from the saved records takes about two minutes and needs neither the datasets nor a GPU. The artifact is organized as follows:

- `code/chronotrace/`: the benchmark library, with the features, adversaries and their training, window estimators, reference graphs, degree controls and defenses;
- `code/scripts/`: data preparation, the experiment runners, the analyses, and the two scripts that generate every macro, table and figure;
- `data/`: download instructions, checksums and the provenance of the dated Epinions build, without the datasets themselves;
- `paper/results/revision/`: one JSON record per target and experiment, the aggregated summaries with their bootstrap intervals and tests, and the log of every experiment batch;
- `tests/`: unit tests on synthetic graphs and end-to-end tests on BitcoinOTC.

Table 23: The submitted evaluation setting: directory releases, in which users left without events remain candidates (mean over 60 targets).

Dataset	Feature set	AUC [95% CI]	AP	P@5	DM-AUC
BitcoinOTC	Degree	0.895 [0.870, 0.919]	0.110	0.13	0.50
	Topology	0.899 [0.874, 0.921]	0.087	0.10	0.49
	Full-R	0.899 [0.875, 0.921]	0.081	0.11	0.52
	Temporal	0.942 [0.930, 0.953]	0.103	0.14	0.78
	Sgn-Struct	0.935 [0.917, 0.950]	0.117	0.12	0.73
	ChronoTrace	0.947 [0.934, 0.958]	0.122	0.18	0.81
	Degree minus label reference	+0.040 [+0.013, +0.064]			
BitcoinAlpha	Degree	0.838 [0.800, 0.873]	0.086	0.08	0.50
	Topology	0.871 [0.843, 0.897]	0.068	0.06	0.54
	Full-R	0.869 [0.842, 0.896]	0.067	0.07	0.55
	Temporal	0.908 [0.885, 0.929]	0.094	0.10	0.80
	Sgn-Struct	0.925 [0.907, 0.941]	0.105	0.12	0.80
	ChronoTrace	0.930 [0.914, 0.946]	0.103	0.11	0.82
	Degree minus label reference	−0.015 [−0.053, +0.021]			
Epinions	Degree	0.830 [0.784, 0.873]	0.048	0.10	0.50
	Topology	0.920 [0.902, 0.936]	0.020	0.02	0.58
	Full-R	0.928 [0.910, 0.943]	0.021	0.04	0.51
	Temporal	0.915 [0.895, 0.934]	0.022	0.04	0.61
	Sgn-Struct	0.915 [0.896, 0.934]	0.019	0.01	0.62
	ChronoTrace	0.927 [0.907, 0.944]	0.021	0.04	0.59
	Degree minus label reference	−0.083 [−0.130, −0.039]			

Table 24: Degree adversary as the candidate pool shrinks (every former neighbor kept; mean over targets).

Dataset	Pool	AUC	AP	P@5	S@10	Best
Epinions (dated)	2,000	0.924	0.275	0.34	0.81	7
	5,000	0.924	0.182	0.25	0.70	16
	10,000	0.924	0.128	0.20	0.57	30
	20,000	0.924	0.090	0.16	0.50	59
	50,000	0.924	0.055	0.11	0.38	147
	all	0.924	0.051	0.10	0.35	167
Epinions (static)	2,000	0.914	0.251	0.33	0.81	15
	5,000	0.914	0.162	0.22	0.72	36
	10,000	0.914	0.111	0.16	0.62	70
	20,000	0.914	0.070	0.11	0.49	140
	50,000	0.914	0.037	0.06	0.33	348
	all	0.914	0.017	0.02	0.22	930

Table 25: Static SNAP Epinions, edge-list release, full candidate universe (60 targets). DM-AUC: degree-matched AUC. Excess: excess over degree.

Feature set	AUC [95% CI]	AP	P@5	DM-AUC	Excess
Degree	0.914 [0.893, 0.934]	0.017	0.02	0.50	0.000
Topology	0.917 [0.896, 0.936]	0.019	0.04	0.54	+0.007*
Full-R	0.917 [0.897, 0.936]	0.022	0.04	0.60	+0.013*
ChronoTrace	0.929 [0.910, 0.946]	0.027	0.04	0.71	+0.038*

* $p < 0.05$, two-sided sign-flip permutation test (unadjusted).

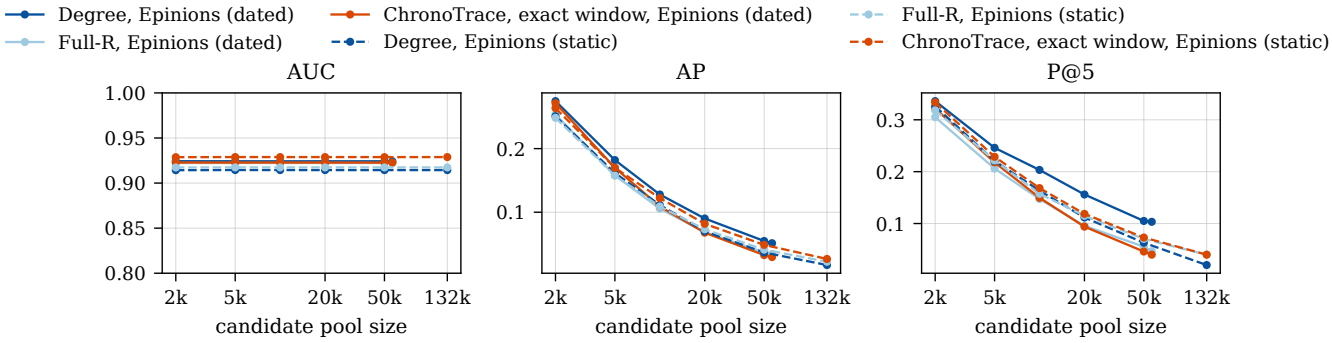


Figure 12: AUC, AP and P@5 as the candidate pool shrinks, with every former neighbor kept.